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RESEARCH ARTICLE

Toward a full-scale model of renal hemodynamics using a reconstructed vascular tree

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Abstract

The kidney's vascular network stands out because *1*) the microcirculation not only supplies the tissues with oxygen and nutrients but also supports glomerular filtration in each nephron, *2*) it contains the tubuloglomerular feedback, a mechanism that contributes to renal blood flow autoregulation and is unique to the kidney, and *3*) the topology of the renal arterial network influences signaling along the vessels mediating nephron-nephron interactions. We have developed a full-scale vascular model of the rat kidney based on a reconstructed vascular network combined with a nephron model that includes glomerular filtration, tubular reabsorption, and autoregulation of afferent arteriolar resistances. The model evaluates the steady-state operating conditions of approximately 30,000 nephrons in a rat kidney and the efficiency of autoregulation under normal and pathological conditions. The simulation results show how the regulated afferent arteriolar resistances stabilize blood flow in the reconstructed full-scale renal vascular network. It is concluded that by using a reconstructed renal vascular tree, it is possible to develop a realistic full-scale model of the regulation of renal hemodynamics as a first step toward creating a virtual kidney.

NEW & NOTEWORTHY We have developed the first full-scale steady-state model integrating a realistic vascular network topology of the kidney and its hemodynamic regulatory mechanisms. The vascular network is combined with approximately 30,000 nephron models that include glomerular filtration, tubular reabsorption, and autoregulation of the afferent arteriolar resistances. By simulating the adaptive properties of the renal microcirculation at steady state, our approach demonstrates the feasibility of utilizing a reconstructed vascular network for comprehensive modeling of renal function.

autoregulation; computational simulation; nephron model; renal blood flow; renal vasculature

INTRODUCTION

Blood vessels in the kidney form a demand-driven, balanced, and connected resource-distribution network. The kidney, like other organs, faces two concurrent challenges with regard to its blood flow. The first is that arterial blood pressure fluctuates predominantly due to daily variation in physical activity (1). Variations in blood pressure cause changes in blood perfusion that are not related to the metabolic demands of the organ and, if large, can damage the organ microcirculation. Autoregulation is the term used for the intrinsic mechanisms in every organ that minimize the impact of pressure variations. The second challenge is to adjust organ perfusion to meet metabolic demands. This requires feedback from the cells of the organ to the upstream vascular segments to adjust their hemodynamic resistances. By adjusting the hemodynamic resistances, the organ can achieve a balance between blood flow and the metabolic demands of the tissues.

In the kidney, the arterial network originates from the renal artery, branches, and forms a rooted tree that reaches all regions of the renal cortex. The terminal arteries at the top of the tree end in pairs or triples of afferent arterioles, where each afferent arteriole supplies a single glomerulus (2, 3). In addition, afferent arterioles may arise directly from the larger branches of the network (3).

The renal microcirculation has been the subject of many mathematical models during the last decades (4–11). The models have investigated the interplay between vascular factors, such as blood pressure and flow, and nephron function, such as glomerular filtration rate and tubular function. The kidney consists of a large number of functional units, nephrons, where the human kidney has ~1 to 1.2 million nephrons and the rat kidney has ~30,000 nephrons. Despite this, most models of renal hemodynamics have focused only on the function of a single nephron, assuming that the overall function of the kidney can be estimated by scaling the results of a single nephron (4, 9, 10, 12). However, several studies have made it clear that individual nephrons do not function as independent units (13). Rather, there is a significant interaction between neighboring nephrons (8). This is due to both the exchange of information along the vessels

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and the effects caused by local changes in the pressure and flow in the vascular network (13).

This has led to an increased interest in developing mathematical models of renal hemodynamics that include several nephrons, which interact through a vascular network (8, 14–17). The results of these studies point to a clear functional significance of the interactions between nephrons. However, the models have only included a relatively small number of interacting nephrons (<32) (8, 14–17).

The reason for the small number of nephrons in the existing models has not only been the lack of computational resources but also the lack of detailed information on the topology of the renal vascular network has been a limiting factor. Methods for imaging the renal vasculature cannot resolve the smallest vessels in the arterial network and therefore cannot provide the information necessary to construct a realistic model of the renal vascular tree.

Motivated by the lack of a realistic full-scale model of the renal vascular tree, Xu et al. (18) have developed a novel hybrid approach to reconstruct a full-scale renal arterial network through a physiologically based constructive optimization method that incorporates anatomical data extracted from a scan of a rat kidney. The result is a full-scale renal arterial tree that starts from the renal artery and ends in the afferent arterioles and has been shown to agree well with existing anatomical data (19).

The aim of the present paper is to investigate the usefulness of the reconstructed vascular network for a full-scale model of the regulation of renal hemodynamics. In the present paper, we construct a steady-state model of renal autoregulation of blood flow based on the reconstructed full-scale model of the renal vascular tree from Ref. 18. To the leaves of the tree (the afferent arterioles) are added mathematical models of nephrons, which include the main mechanisms of renal autoregulation, that is, the tubuloglomerular feedback mechanism and the myogenic response. With around 30,000 nephrons, the model is the first full-scale steady-state model of blood flow regulation in a rat kidney and it represents a first step toward the creation of a virtual kidney. By integrating a realistic vascular network topology with individual models of nephron autoregulatory mechanisms, the model allows for an exploration of adaptation and optimization within the nephron-vascular network and enables, through simulation, a study of both normal and pathological conditions within the vascular network.

MATERIALS AND METHODS

The mathematical model includes four submodels: a glomerular model, a tubular model, a model of the afferent arteriole, and the full-scale model of the renal arterial tree, as shown in Fig. 1. The nephron model (glomerular, tubular,

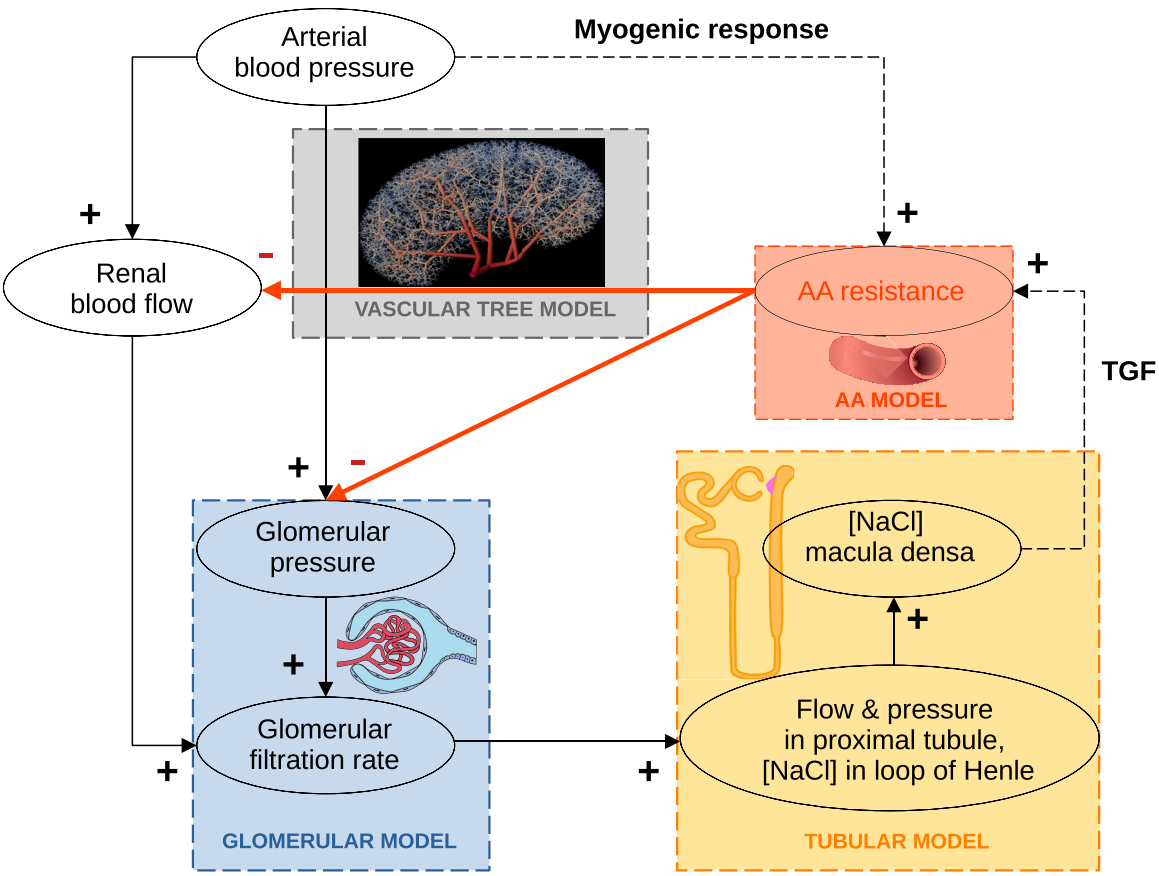


Figure 1. An overview of the full-scale model of the nephron-vascular network and the simulation of renal autoregulation. The full model consists of four submodels: a glomerular model, a tubular model, an afferent arteriolar (AA) model, and a vascular tree model. The arterial blood pressure serves as the input to the model. The model integrates the mechanisms responsible for autoregulation of renal blood flow: the myogenic response and the tubuloglomerular feedback mechanism (TGF).

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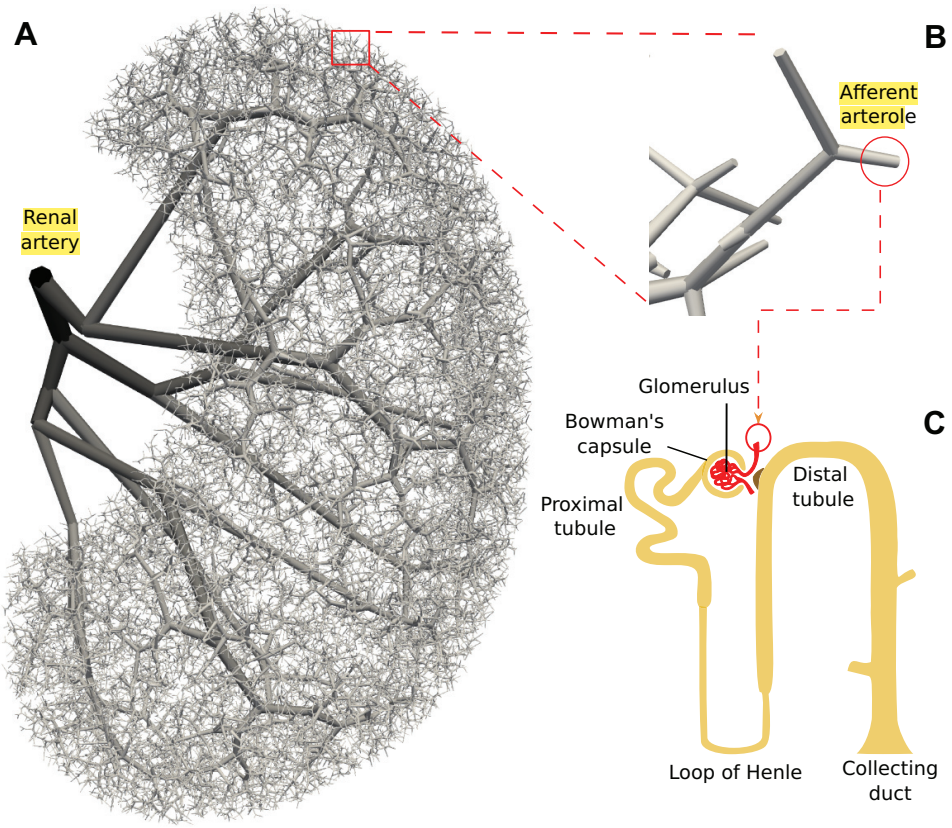


Figure 2. Illustration of the topology of the reconstructed vascular tree (*A*) where each leaf (afferent arteriole) (*B*) is connected with a nephron model as the feeding vessel to the glomerulus (*C*). Together they constitute a model of the full-scale nephron-vascular network. Note that each of the terminals in *B* is connected with the nephron model depicted in *C*, but only one is plotted for illustrative reasons.

and afferent arteriolar models) is adapted from a previously published model (6), whereas the full-scale renal arterial network is reconstructed using a hybrid approach described in Ref. 18. This section provides an overview of the essential components of the model, with detailed explanations available in APPENDIXES A and B.

Topology of Full-Scale Renal Arterial Network

The main contribution of this work is the attachment of nephrons to the leaves (afferent arterioles) of a full-scale renal arterial network, which is reconstructed using a hybrid approach using both the geometries extracted from a micro-CT scan of a rat kidney and a physiologically based generative model as described in Ref. 18. The output of the hybrid method is a full-scale renal arterial network that starts at the renal artery and ends at the afferent arterioles, as shown in Fig. 2*A*. This arterial network can be represented as a tree [a directed acyclic graph $G \equiv (V, E)$], where the edges (E) correspond to individual vessels, connecting at the branch points (nodes), V, and where the top nodes correspond to the end of the afferent arterioles. Each single vessel in the tree is assumed to have a uniform radius throughout its length.

The leaves (the top branches) represent the afferent arterioles, whereas the single root is the renal artery. All intermediate arteries, for example, interlobular and arcuate arteries, correspond to the internal edges of the tree structure. A method of labeling the branches of a tree, for example, a vascular network, is Strahler ordering (19–21). It begins by labeling all leaves (the afferent arterioles in this case) as having Strahler order 0, with an increasing order assigned to the parent vessels at the branch points until it reaches the root (the renal artery in this case) of the tree, as shown in Fig. 3*A*. In Ref. 18, we showed that the reconstructed full-scale renal arterial tree is in close agreement with existing anatomical data (19) with respect to different

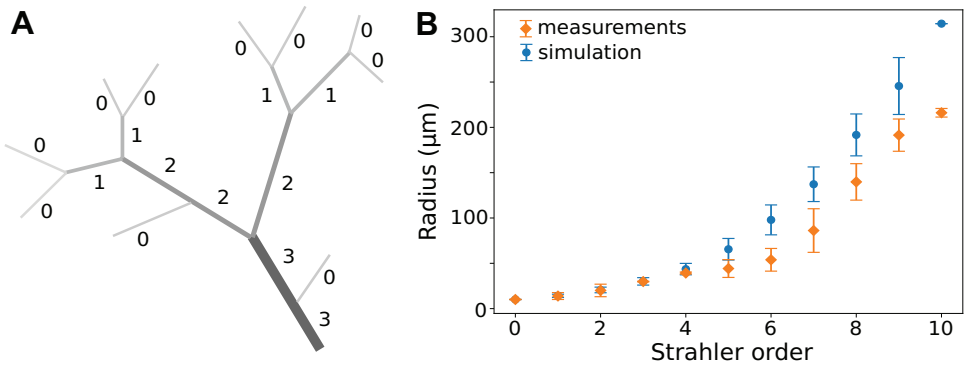


Figure 3. Illustration of the principle of Strahler ordering (*A*) and an example of a Strahler order-related morphological feature of the reconstructed vascular tree, the means and SD of the radii of vessels as a function of their Strahler order (*B*). In the reconstructed vascular tree, all afferent arterioles correspond to leaves and have Strahler order 0, whereas the single renal artery is the root and has Strahler order 10. Note that for simplicity, *A* only shows vessels up to order 3.

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Strahler-order-related morphometric parameters. As an example, Fig. 3*B* compares the mean and SD for the radii of the vessels at the respective Strahler orders in the reconstructed vascular tree with the experimental results obtained in a cast of a rat vascular tree. For additional comparisons, see Ref. 18.

With the topology of the renal vascular tree reconstructed using the method from Ref. 18, the functional units—nephrons (Fig. 2*C*)—are added to the afferent arterioles (Fig. 2*B*), which together form the full-scale nephron-vascular network. The reconstructed tree from Ref. 18 involves 29,608 leaves (afferent arterioles), and thus, 29,608 nephron models are coupled to the renal arterial tree.

The Mathematical Model

Glomerular model.

As blood flows from the afferent arterioles through the glomerular capillaries, the fluid is filtered through the filtration barrier, which is impermeable to plasma proteins. Protein is conserved within the capillaries, and the glomerular filtration rate in a single nephron (SNGFR) can be calculated from the product of the fractional change in protein concentration in the glomerular capillaries, $\frac{C_A}{C_E}$, and the plasma flow entering the capillaries, Q_A (6):

$$\mathrm{SNGFR} = \left(1 - \frac{C_A}{C_E}\right) Q_A \tag{1}$$

where C_A is the plasma protein concentration at the start and C_E is the concentration at the end of the glomerular capillaries.

The filtration process that causes the change in the protein concentration, C, along the capillary is proportional to the local pressure difference:

$$\frac{dC}{dx} = \frac{K_f}{L Q_A C_A}(C^2)[P_{GC} - P_T(0) - \pi(C)] \tag{2}$$

where x is the position along the glomerular capillary, K_f is the filtration coefficient, and L is the length of the capillary. P_{GC}, $P_T(0)$, and $\pi(C)$ are the capillary hydrostatic pressure, the tubular pressure, and the colloid osmotic pressure due to plasma proteins, respectively. $\pi(C)$ is a nonlinear function that can be approximated by fitting a second-order polynomial to experimental data

$$\pi(C) = aC + bC^2 \tag{3}$$

where a and b are empirical constants (22).

Plasma flow entering the capillaries, Q_A, and capillary hydrostatic pressure, P_{GC}, are recovered from the full-scale renal arterial tree model, as described in *Full-scale renal arterial tree model*.

The initial condition necessary to solve *Eq. 2* is $C(0) = C_A$, the concentration of proteins in arterial plasma. From this, the concentration $C_E = C(L)$ at the end of the capillary ($x = L$) can be calculated. For parameter details, see Table 1.

Tubular model.

In the tubular model, the differential equations for the flow and the NaCl concentration are based on local conservation of mass. The NaCl concentration in the tubular fluid (C_S) changes because the tubule reabsorbs NaCl (J_S) and/or water (J_V). Since the flow of tubular fluid has a low Reynold's number, it is assumed that the pressure drop within the tubule is given by Poiseuille's law. Thus, the total behavior is expressed by a system of three coupled differential equations consisting of pressure, P_T, flow, Q_T, and NaCl concentration, C_S:

$$\begin{cases} \dfrac{dP_T}{dz} = -\dfrac{8\mu}{\pi r^4} Q_T, \; 0 < z < Z \\ \dfrac{dQ_T}{dz} = -J_V, \; 0 < z < Z \\ \dfrac{dQ_T C_S}{dz} = Q_T \dfrac{dC_S}{dz} + C_S \dfrac{dQ_T}{dz} = -J_S, \; 0 < z < Z \end{cases} \tag{4}$$

where Z represents the total length of the tubule.

The present tubular model only includes the initial two segments: the proximal tubule and the loop of Henle, where the latter is divided into a descending and an ascending limb. The expressions for J_V and J_S are defined differently based on the relevant tubular segment (see APPENDIX A). The objective in solving the tubular model is to compute the NaCl concentration at the macula densa, $C_S(md)$, which is located in the ascending limb of the loop of Henle. The filtration rate of the glomerular model is equal to the flow of tubular fluid at the beginning of the tubule, thus $\mathrm{SNGFR} = Q_T(0)$, where $Q_T(0)$ serves as a boundary condition to *Eq. 4*, as shown in Fig. 1. Details on the solution of the three coupled differential equations are given in APPENDIX A. For parameter details, see Table 2.

Afferent arteriole model.

The tension in the wall of the afferent arteriole is given by Laplace's equation:

$$P_v r_v = T \tag{5}$$

where the transmural pressure P_v is calculated as the average of the pressures at each end of the vessel segment, and r_v is the inner radius of the afferent arteriole.

The tension, T, is assumed to be a sum of two components: an elastic term, T_e, stemming from the passive stretch of the elastic elements in the vessel wall, and an active tension, T_m, generated by the contraction of the vascular smooth muscle:

Table 1. *Glomerulus parameters*

Parameter	Value	Unit	Explanation	Ref.
C_A	57	$g \cdot L^{-1}$	Plasma protein concentration in afferent arteriole	(23)
Ht_A	0.5	N/A	Hematocrit in afferent arteriole	
K_f	2.5	$nl \cdot min^{-1} \cdot mmHg^{-1}$	Filtration coefficient	(23)
R_E	0.209	$mmHg \cdot min \cdot nL^{-1}$	Resistance of the efferent arteriole	(24)
R_{PC}	0.0702	$mmHg \cdot min \cdot nL^{-1}$	Resistance of postcapillary vessels	(24)
a	0.1631	$mmHg \cdot L \cdot g^{-1}$	Constant for colloid osmotic pressure, $\pi(C)$	(22)
b	0.00294	$mmHg \cdot L^2 \cdot g^{-2}$	Constant for colloid osmotic pressure, $\pi(C)$	(22)

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Table 2. *Tubule parameters*

Parameter	Value	Unit	Explanation	Ref.
K_m	20	mM	NaCl affinity for active reabsorption	(25)
V_{max}, pre-MD	1.2×10^{-7}	$mmol \cdot cm^{-1} \cdot s^{-1}$	NaCl maximum active reabsorption rate	(25)*
V_{max}, post-MD	0.78×10^{-7}	$mmol \cdot cm^{-1} \cdot s^{-1}$	NaCl maximum active reabsorption rate	(25)*
L_V	2×10^{-5}	$cm^2 \cdot L \cdot osmol^{-1} \cdot s^{-1}$	Water permeability in descending limb	(26)
n_S	2	$mol \cdot osmol^{-1}$	Number of osmoles per mole of solute	
L_S	3.4×10^{-7}	$cm^2 \cdot s^{-1}$	NaCl permeability in descending limb	(26)
L_S, short	3.4×10^{-7}	$cm^2 \cdot s^{-1}$	NaCl permeability in ascending limb	(26)
L_S, intermediate	5.2×10^{-7}	$cm^2 \cdot s^{-1}$	NaCl permeability in ascending limb	(26)*
L_S, long	5.6×10^{-7}	$cm^2 \cdot s^{-1}$	NaCl permeability in ascending limb	(26)*
α	1.65×10^{-2}	$(nL \cdot min^{-1} \cdot mmHg^{-5})^{1/4}$	Empirical constant, tubular pressure	(27)
β	0.924	$(nL \cdot min^{-1} \cdot mmHg^{-1})^{1/4}$	Empirical constant, tubular pressure	(27)
θ	1.3	cm^{-1}	Length constant, proximal tubular reabsorption	(27)
κ	33.6	$nL \cdot min^{-1} \cdot cm^{-1}$	Maximum rate, proximal tubular reabsorption	(27)
r_{tubule}	12×10^{-4}	cm	Inner radius of proximal tubule	(28)
r_{henle}	10×10^{-4}	cm	Inner radius of loop of Henle	(25)

*Slightly adjusted so that the model produces an average $C_S(md)$ of ~45 mM in the control case. MD, macula densa.

$$T = T_e + T_m \tag{6}$$

where

$$T_e = \sigma_e h(r_v)\left(\exp\left(k_e[r_v - r_e]\right) - 1\right) \tag{7}$$

and

$$T_m = \sigma_m h(r_v) f(x_{myo}) \exp\left(-k_m[r_v - r_m]^2\right). \tag{8}$$

The term $f(x_{myo})$ (*Eq. 10*) expresses the degree of activation of the vascular smooth muscle cells of the afferent arteriole as a function of the error signal from the myogenic mechanism, x_{myo} (*Eq. 11*). The thickness of the wall, $h(r_v)$, varies with the radius so that the cross-sectional area of the vessel wall remains constant. Details of the definition and computation of $h(r_v)$ are given in APPENDIX A. The remaining terms are empirical constants, whose values are given in Table 3.

Combining *Eqs. 5* and *6* provides a relation between the intravascular pressure P_v, the afferent arteriolar radius r_v, and the active T_m and elastic T_e tensions of the vessel wall.

$$P_v r_v = T_e + T_m. \tag{9}$$

$f(x_{myo})$ in T_m (*Eq. 8*) is a sigmoid function, defined as

$$f(x_{myo}) = \frac{3\exp(x_{myo})}{\exp(x_{myo}) + 2\exp(-0.5\,x_{myo})}. \tag{10}$$

Myogenic response. The degree of activation of vascular smooth muscle cells arising through the myogenic mechanism, x_{myo}, is given by

$$x_{myo} = G\left(T - T_0\left(1 - x_{tgf}\right)\right) \tag{11}$$

where G is the gain of the myogenic mechanism and T_0 is the set point. The set point is modified by the signal from the tubuloglomerular feedback mechanism, x_{tgf} (*Eq. 12*).

Tubuloglomerular feedback. The action of tubuloglomerular feedback is assumed to be mediated solely through the afferent arteriole (6) and to be described by a logistic equation that relates the feedback signal, x_{tgf}, to the NaCl concentration at the macula densa, $C_S(md)$:

$$x_{tgf} = \xi_{max} - \frac{\psi}{1 + \exp\left[k\left(C_S(md) - C_{1/2}\right)\right]} \tag{12}$$

where $C_S(md)$ is given by *Eq. 4* in the tubular model.

Note that in the model, the effect of TGF is indirect, acting by modifying the set point of the myogenic mechanism, see *Eq. 11*. A simple additive model, in which smooth muscle activation given by $x_{myo} + x_{tgf}$, results in antagonism between the two mechanisms, thereby impairing autoregulation (30).

Full-scale renal arterial tree model.

With a full-scale vascular tree model that starts from a single renal artery and ends in the afferent arterioles (18), the hemodynamic resistance in each vessel segment connecting nodes *i* and *j* is computed by Poiseuille's law (assuming laminar flow):

Table 3. *Afferent arteriole parameters*

Parameter	Value	Unit	Explanation	Ref.
h_0	2	μm	Afferent arteriole reference wall thickness	(29)
r_0	10	μm	Afferent arteriole reference radius	(29)
σ_e	25.56	$dyn \cdot cm^{-2}$	Elasticity coefficient	(29)
k_e	17,304	cm^{-1}	Elasticity coefficient	(29)
r_e	7	μm	Set point for elastic response	(29)
σ_m	7.23×10^5	$dyn \cdot cm^{-2}$	Muscle contraction coefficient	(29)
k_m	8.75×10^6	cm^{-2}	Muscle contraction coefficient	(29)
r_m	12.5	μm	Set point for muscle contraction	(29)
G	0.06	$cm \cdot dyn^{-1}$	Gain of the myogenic response	(29)
T_{T0}	81.13	$dyn \cdot cm^{-1}$	Set point for wall tension	(29)*

*Slightly modified to match the initial conditions to the present model (see Table 4).

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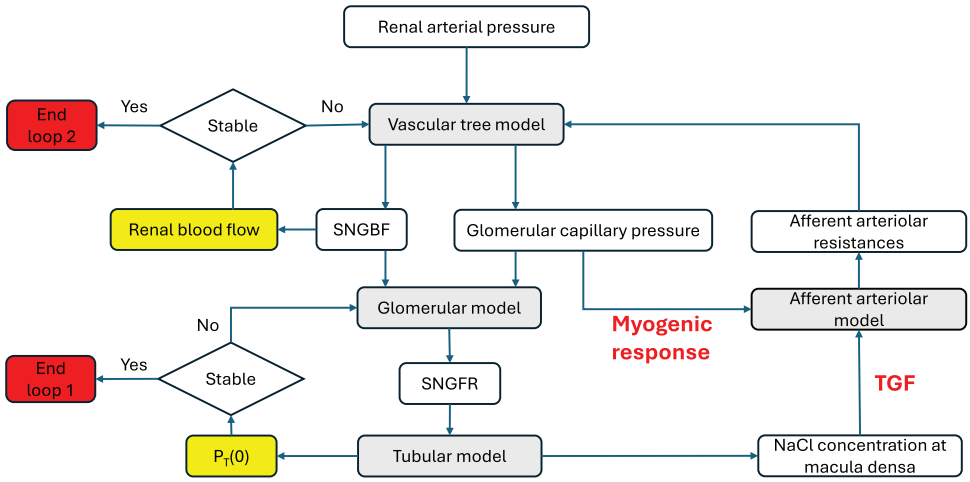


Figure 4. Flowchart for solution of the model. The renal arterial pressure together with an initial value for the afferent arteriolar resistances serve as the inputs to the vascular tree model. The vascular tree model provides the single nephron blood flow (SNGBF) and the glomerular capillary pressure. These are the input to the glomerular model. Given a guess for the pressure at the start of the proximal tubule, $P_T(0)$, the glomerular model will give as output the single nephron glomerular filtration rate SNGFR $= Q_T(0)$, which serves as the initial flow value to the tubular model. The tubular model will give as output the NaCl concentration at the macula densa, which is the stimulus for TGF, and the pressure in the proximal tubule, $P_T(z)$, including a new value for $P_T(0)$. The new value for $P_T(0)$ allows a recalculation of the glomerular model. This loop is repeated until $P_T(0)$ has converged to a stable value (*loop 1*). Following the convergence of the first loop, new values for the afferent arteriolar resistances are calculated using the afferent arteriolar model. These are determined by the myogenic response and TGF. The myogenic response is induced by the pressure at the beginning and end (glomerular capillary pressure) of the afferent arteriole. TGF depends on the NaCl concentration at the macula densa. With a new set of values for the afferent arteriolar resistances, the full-scale arterial tree will provide a new set of single nephron blood flows SNGBF and glomerular capillary pressures. This process will be iterated until convergence of the renal blood flow (*loop 2*), which is the sum of the SNGBF in the individual nephrons. The gray boxes refer to the four submodels, and the yellow boxes to the iterated variables.

$$R_{i,j} = \frac{8\mu l_{i,j}}{\pi r_{i,j}^4} \tag{13}$$

where $l_{i,j}$ is the length of the segment and $r_{i,j}$ is the inner radius of the vessel.

For any intermediate node n in the tree, Kirchhoff's current law (first law) states that inflow equals the sum of outflows:

$$Q_{p,n} = \sum_{i \in B_v/p} Q_{n,i} \tag{14}$$

where B_n and p denote the set of neighboring nodes of n and the single parent node of n, respectively.

Combined with Poiseuille's law

$$Q_{i,j} = \frac{P_i - P_j}{R_{i,j}}. \tag{15}$$

Kirchhoff's current law can then be rewritten as

$$\frac{P_p - P_n}{R_{p,n}} = \sum_{i \in B_v/p} \frac{P_n - P_i}{R_{n,i}}. \tag{16}$$

The above equation can be rearranged to isolate P_n:

$$\left(\frac{1}{R_{p,n}} + \sum_{i \in B_v/p} \frac{1}{R_{n,i}}\right) P_n - \left(\frac{P_p}{R_{p,n}} + \sum_{i \in B_v/p} \frac{P_i}{R_{n,i}}\right) = 0. \tag{17}$$

With appropriate boundary conditions, the above equation can be solved to give the pressures, and thus the flows, in all nodes of the vascular tree; see APPENDIX B for details.

Simulation Pipeline

Figure 4 illustrates the flowchart of the simulation that integrates the four models and solves them consecutively.

The simulation is started from a prebuilt full-scale arterial tree model $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ that includes an initial value for the resistances of the afferent arterioles together with the fixed resistances for the larger vessels (for initial value details, see Table 4). Given the boundary condition of the arterial pressure P_{art}, Kirchhoff's current law is used to calculate the pressure at each node and the blood flow in each vessel of the tree, as discussed in *Full-scale renal arterial tree model*.

The calculated blood flows and pressures at the ends of the afferent arterioles are the input to the Glomerular model. Specifically, the blood flow in the afferent arteriole is equal to the glomerular blood flow, SNGBF, which is used to compute the plasma flow Q_A by

$$Q_A = (1 - Ht_A) \times \text{SNGBF} \tag{18}$$

where Ht_A is a constant representing the hematocrit in the arterial blood.

With plasma flow Q_A and capillary hydrostatic pressure P_{GC} (equivalent to the pressure at the end of the corresponding afferent arteriole) given, the only unknown in the *Glomerular model* is the tubular pressure $P_T(0)$ (pressure at the beginning of the tubule). Given an initial guess of $P_T(0)$, the filtration rate SNGFR is calculated as discussed in the *Glomerular model*.

SNGFR from the glomerular model serves as a boundary condition ($Q_T(0) =$ SNGFR) to the tubular model. The tubular model is then solved for the NaCl concentration at the macula densa $C_S(\text{md})$, as well as for a new value for the tubular pressure $P_T(0)$. The new value for tubular pressure is then used to solve the glomerular model (see *Glomerular model*), obtaining a new value for SNGFR for use in the tubular model. This whole process is iterated until $P_T(0)$ converges (*loop 1*). Since

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Table 4. *Initial conditions*

Parameter	Value	Unit	Explanation
r_v	10.1 ± 0.14	μm	Radius of afferent arteriole
P_{GC}	58.0 ± 4.5	mmHg	Glomerular capillary pressure
Q_A	117.0 ± 9.0	nL · min^{-1}	Afferent arteriolar plasma flow
T	81.1 ± 5.8	dyn · cm^{-1}	Wall tension
$Q_T(0)$	39.0 ± 3.0	nL · min^{-1}	Glomerular filtration rate

Means ± standard deviation. The distribution of the afferent arteriolar radius r_v stems from the reconstructed vascular tree. The remaining parameter distributions are obtained by solving the model in the control case without autoregulation.

each nephron is independent, *loop 1* can be done in parallel for all nephrons.

The NaCl concentration at the macula densa, $C_S(md)$, at the convergence of $P_T(0)$ is used in *Eq. 12* to calculate tubuloglomerular feedback, which is the input to the afferent arteriole model to calculate the myogenic response x_{myo}, allowing calculation of active T_m and elastic T_e tensions. The calculated tensions are used in *Eq. 9* to solve the radii of the afferent arterioles, r_v.

The adjusted afferent arteriolar radii induce a new set of resistances for the afferent arterioles, *Eq. 13*. Kirchhoff's law (*Eq. 2.2*) is then reapplied on the full-scale vascular tree with the new resistances for the afferent arterioles, resulting in a new set of flows SNGBF and pressures P_{GC} for the afferent arterioles. Then, these are used as input to the glomerular model, and the whole process is repeated until renal blood flow converges (*loop 2*).

In short, the process of solving the model involves two iteration loops. The first loop is used to solve for $P_T(0)$ and $C_S(md)$ for the given values of the resistances of the afferent arterioles. At the convergence of the first loop for $P_T(0)$, the value obtained for $C_S(md)$ is then used to calculate new values for the resistances of the afferent arterioles, and blood flow and pressures in the vascular tree. The second loop is repeated until renal blood flow converges. Notice that the first loop is embedded within the second loop.

Detailed hardware environment and software packages used in the whole simulation are presented in APPENDIX C.

Subpopulations of Nephrons

The nephrons of the renal cortex are not identical. Traditionally, three types of nephrons are described and labeled superficial, midcortical, or juxtamedullary based on the location of their glomerulus within the renal cortex (31, 32), as shown in Fig. 5. Superficial nephrons have glomeruli located near the surface of the kidney with short loops of Henle. Juxtamedullary nephrons have a glomerulus near the border between the cortex and the medulla and have long loops of Henle that extend deep into the medulla. The midcortical nephrons have loops of Henle with intermediate lengths (31, 32). The length of Henle's loop varies between nephrons within each subpopulation. However, for simplicity, we assumed that the lengths of the loops were the same for all nephrons within a given subpopulation, varying only between different subpopulations. In the present model, the renal cortex is assumed to have a depth of 2 mm (18). The nephrons located in the outer 1 mm of the cortex are assumed to be superficial nephrons, which constitute 50% of all nephrons. The midcortical nephrons have their glomeruli in a band of 1 mm to 1.4 mm from the surface of the kidney and constitute 20% of all nephrons. Finally, juxtamedullary nephrons have their glomeruli located from 1.4 mm from the surface and down to the start of the medulla (2 mm from the surface). They constitute 30% of the total nephron population.

TGF has been shown to be stronger in the juxtamedullary compared with the superficial nephrons (33, 34). Hence, we have adjusted the TGF parameters accordingly (see Table 5). Details on the morphology and parameters of the three nephron populations are given in Table 6.

Autoregulation in Single Nephron Model

In the literature, autoregulation of renal blood flow has been widely studied using single nephron models (4, 7, 9–11, 30). In these models, the vasculature from the renal artery to the afferent arteriole is modeled as a single segment, which is then connected in series to the afferent arteriole. This segment represents the larger preafferent arteriolar vessels. The segment is assumed to be unaffected by the autoregulatory process; therefore, its resistance, R_{pre}, is assumed to be constant. The afferent arteriole is the only segment where the autoregulatory process occurs, and its resistance, R_A, will change as a function of pressure, as discussed in *Afferent arteriole model*.

To compare the full-scale model to a single nephron model, autoregulation curves were also computed for three selected single nephrons operating at three different glomerular capillary pressures (see below). The preafferent arteriolar resistance, R_{pre}, of the three nephrons was calculated from the full-scale model using an arterial pressure, P_{art}, of 100 mmHg as:

$$R_{pre} = \frac{P_{art} - P_i}{Q_i} \quad (19)$$

where P_i is the pressure in the node preceding the afferent arteriole of the respective nephrons and Q_i is the corresponding flow in the afferent arteriole calculated from the full-scale model. The autoregulatory response of the afferent arteriole is calculated using the same approach as outlined in *The Mathematical Model* but without the need for a full-scale renal arterial tree model.

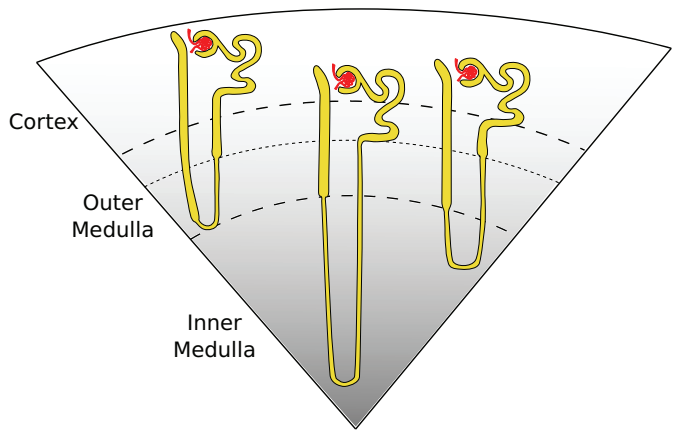


Figure 5. Anatomical features of the three nephron subpopulations. The glomerulus is indicated by the filled red circles. The dotted line indicates the border between the outer and inner stripes of the outer medulla.

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Table 5. *TGF response parameters for the three subpopulations of nephrons*

Parameter	Superficial	Midcortical	Juxtamedullary	Unit	Explanation	Ref.
k	0.085	0.07	0.06	$L \cdot mmol^{-1}$	Sensitivity, TGF	(35)
ψ	0.182	0.2184	0.2583	N/A	Dynamic range, TGF	(25)
ξ_{max}	0.091	0.1183	0.1456	N/A	Maximal TGF response	(25)
$C_{1/2}$	44	44	44	mM	Inflection point, TGF	(36)

References are for superficial and juxtamedullary nephrons; values for midcortical nephrons are interpolated.

RESULTS

Single Nephron: Effects of the Regulatory Mechanisms on the Afferent Arteriolar Radius

To illustrate the effects of the two regulatory mechanisms in the model, Fig. 6 shows the radius of the afferent arteriole for different afferent arteriolar pressures and NaCl concentrations at the macula densa, $C_S(md)$, in a single superficial nephron. Fig. 6*A* (*top left*) shows the TGF response when the pressure in the glomerular capillaries is maintained constant at a value of 57 mmHg. As NaCl concentration ($C_S(md)$) increases, there is a decrease in the radius of the afferent arteriole (the TGF response). Likewise, when the pressure in the glomerular capillaries increases, whereas $C_S(md)$ remains constant at 45 mM [Fig. 6*B* (*top right*)], there is also a decrease in the radius of the afferent arteriole (the myogenic response). Fig. 6*C* (*bottom*) shows that when both pressure and $C_S(md)$ are changed simultaneously, the two regulatory mechanisms act together to regulate the radius of the afferent arteriole.

Vascular Tree: Effect of the Myogenic and TGF Responses

This section describes the behavior of the full-scale vascular tree model with and without myogenic and TGF responses, that is, with and without the autoregulatory mechanisms. In the following simulations, all nephrons are of the superficial type. In the case without myogenic and TGF responses, the radius of all vessels in the tree is generated by Ref. 18. When the input pressure is $P_{art} = 100$ mmHg, the full-scale arterial tree model shows a wide range of values for blood flows in the afferent arterioles and for glomerular capillary pressures, Fig. 7 (blue bars). Although the means (3.9 $nL \cdot s^{-1}$ and 58 mmHg) are in the normal ranges for glomerular capillary pressures and flows (37–39), the distributions are quite broad, with too large or too small values associated with some afferent arterioles. However, without autoregulation, the radius distribution for the afferent arterioles is quite narrow with a mean of 10 μm.

When the autoregulatory mechanisms are active, it is clear that the pressure and flow distributions become narrower, whereas the distribution for the radii is broadened (Fig. 7). Thus, the nephron-vascular network adjusts the flow and pressure in each afferent arteriole by adjusting its radius, through the action of the two feedback mechanisms. In nephrons that initially had low afferent arteriolar blood flow and glomerular capillary pressure, there is a vasodilation of the afferent arterioles, resulting in an increase in pressure and flow, and vice versa in nephrons with high pressures and flows. However, there is only a minimal effect on the mean values of the afferent arteriolar flow and radius and the glomerular capillary pressure.

Figure 8 shows scatter plots of the glomerular capillary pressures in individual nephrons as a function of their position within the vascular tree. It illustrates the importance of the location within the vascular tree for the glomerular capillary pressures. The farther from the root, measured either by distance or number of vascular branches, the lower the pressure. This is expected since hemodynamic resistance increases with the length of the vasculature. The colored crosses labeled n1–n5 in the scatter plots indicate 5 selected nephrons out of the full set of 29,608 nephrons based on their glomerular capillary pressures in the renal arterial tree [from high (n1) to low (n5) pressures].

Figure 9 shows scatter plots for various variables in the full-scale model in the control case with active autoregulation. To better appreciate the combined state of the nephrons, the colored crosses in the four panels indicate the five different nephrons operating under different glomerular capillary pressures in the vascular tree, see Fig. 8. The radius-pressure relationship (Fig. 9*A*) shows that in nephrons with low filtration pressure, the afferent arterioles are dilated, whereas those with high pressure are contracted.

The tension-pressure relationship (Fig. 9*B*) shows that in nephrons with low glomerular capillary pressures, there is a direct positive relationship between pressure and wall tension (T); at these low pressures, the vessels are maximally dilated (see Fig. 9*A*), and therefore, the wall tension will be

Table 6. *Length of tubule segments for the three subpopulations of nephrons*

Tubule Segment	Superficial	Midcortical	Juxtamedullary	Unit	Ref.
Proximal tubule	1	1	1	cm	(32)
Thin descending limb	0.3	0.55	0.8	cm	(32)
Thin ascending limb	0	0.25	0.5	cm	(32)
Thick ascending limb (medullary)	0.3	0.3	0.3	cm	(32)
Thick ascending limb (cortical)	0.2	0.1	0.02	cm	(32)
Thick ascending limb (post-MD)	0.1	0.1	0.1	cm	(32)
Transition zone	0.05	0.05	0.05	cm	(32)

In all cases, the macula densa is assumed to be located close to the end of the cortical part of the thick ascending limb followed by a short post macula densa (MD) part of the thick ascending limb before entering the distal convoluted tubule. References are for superficial and juxtamedullary nephrons, values for midcortical nephrons are interpolated.

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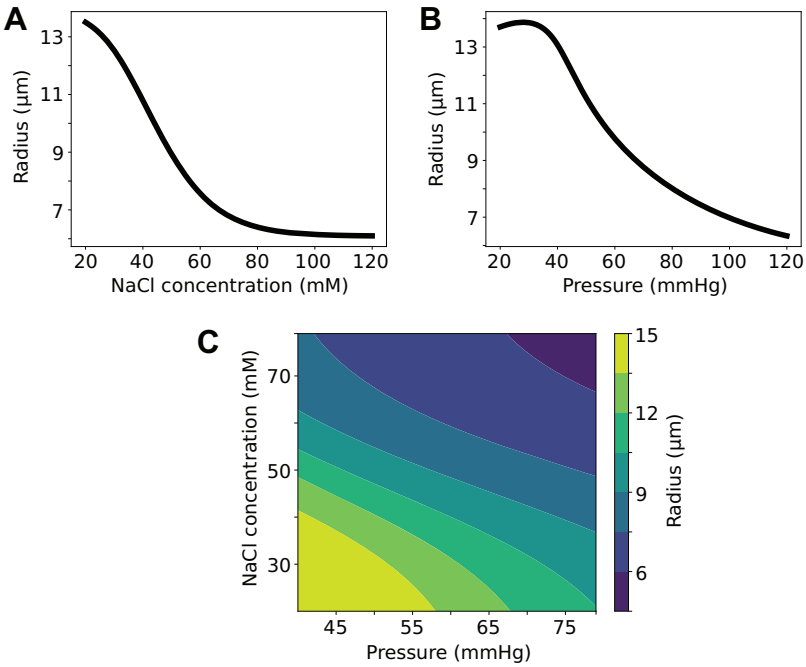


Figure 6. The single superficial nephron model reproduces the well-known dependency of the radius with increasing NaCl concentration at the macula densa, when the pressure in the glomerular capillaries is held constant at 57 mmHg (*A*) and with increasing arterial pressure, when the $C_S(md)$ is held constant at 45 mM (*B*). *C*: the resulting radius (color bar, μm) as a function of both the $C_S(md)$ and the pressure.

positively correlated with the pressure according to the Laplacian equation (*Eq. 5*). At higher pressures, the vessel contracts and the reduction in radius is relatively greater than the increase in pressure, so the net result is a decrease in wall tension with increasing pressure.

Fig. 9*C* shows the TGF signal (x_{TGF}) versus the NaCl concentration at the macula densa $C_S(md)$. The relationship is sigmoidal, as would be expected from the TGF function in individual nephrons (see *Eq. 12*). Some nephrons have reached the maximum value for the TGF mechanism. Fig. 9*D* shows that the TGF and myogenic responses are positively correlated, so they act together to provide effective autoregulation.

Vascular Tree: Autoregulation

Full-scale arterial tree model.

When the arterial pressure varies from the control value of 100 mmHg, the renal autoregulatory mechanisms minimize the change in renal blood flow by adjusting the afferent arteriolar resistances. Here, we simulate autoregulation in the full-scale vascular model by varying arterial pressure from 80 mmHg to 200 mmHg and calculating the resulting renal blood flow with and without autoregulation. In these simulations, all nephrons are of the superficial type.

Figure 10 shows the resulting renal blood flow with and without autoregulation. Without autoregulation (blue curve), renal blood flow increases linearly with increasing renal arterial pressure. In contrast, the autoregulation provided by TGF and the myogenic response is effective in minimizing the increase in renal blood flow when arterial pressure increases from 80 mmHg to 200 mmHg (orange curve). Autoregulation saturates when the arterial pressure becomes too high, as can be seen from the larger slope at arterial pressures from 180 mmHg to 200 mmHg.

We simulate the effect of blocking TGF (which can be achieved experimentally by giving a loop diuretic) so that only the myogenic response is active. Note that in the model, TGF acts by modifying the set point of the myogenic response (see *Eq. 11*), so it is not possible to simulate the TGF response alone without the myogenic response. If TGF is inactivated, only the myogenic response contributes to autoregulation and cannot fully compensate for the increased blood pressure (green curve). It is evident that both mechanisms are necessary for efficient autoregulation of renal blood flow.

Figure 11 shows the distributions of blood flow, afferent arteriolar radius, and tubular pressure and flow at the end of the thick ascending limb of the loop Henle when renal arterial pressure, P_{art}, increases from 100 mmHg to 140 mmHg. Increasing arterial pressure results in a decrease in the radii of the afferent arterioles (Fig. 11*A*). Note that the system begins to saturate. Several nephrons, represented by the left peak in the red histogram, reach a lower limit for the radii of their afferent arterioles. The blood flow in the nephrons increases slightly (Fig. 11*B*), and the beginning saturation of the autoregulatory response is evident from the long right tail in the blood flow distribution at 140 mmHg. As a result of autoregulation, renal blood flow is only slightly increased from 6.9 $mL \cdot min^{-1}$ at 100 mmHg to 7.55 $mL \cdot min^{-1}$ at 140 mmHg. At 100 mmHg, both the tubular pressure and the flow rate at the end of the thick ascending limb of the loop of Henle show narrow distributions with medians of 8.5 mmHg (8.2 mmHg; 8.8 mmHg) (Fig. 11*C*) and 0.18 $nL \cdot s^{-1}$ (0.17 $nL \cdot s^{-1}$; 0.20 $nL \cdot s^{-1}$) (Fig. 11*D*), where the numbers in brackets are the 25% and 75% quantiles. At 140 mmHg, there is an increase in the median values of tubular flow and pressure to 10.0 mmHg and 0.24 $nL \cdot s^{-1}$, respectively. The distributions are broadened with a pronounced rightward tail due to saturation of the autoregulatory mechanisms, (9.2 mmHg; 11.9 mmHg) and (0.21 $nL \cdot s^{-1}$; 0.31 $nL \cdot s^{-1}$).

To quantitatively assess the efficiency of autoregulation, the autoregulation index (ARI) was calculated at different levels of arterial pressure. The ARI is given as the ratio between

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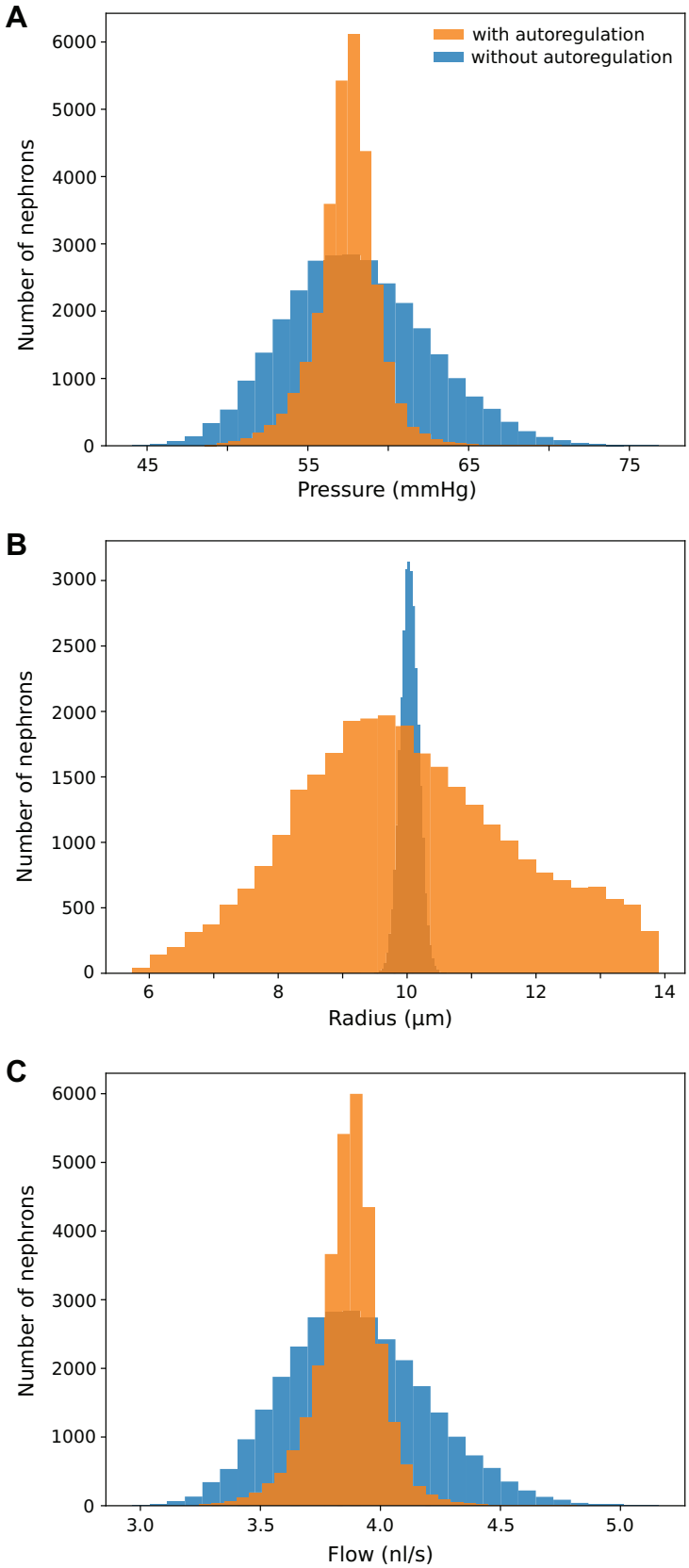


Figure 7. Histograms showing the effect of the autoregulatory mechanisms on the distributions of glomerular capillary pressure (*A*), afferent arteriolar radius (*B*), and afferent arteriolar blood flow (*C*) without (blue) and with (orange) autoregulation when the renal arterial pressure $P_{art} = 100$ mmHg (all nephrons are of superficial type). Renal blood flow (RBF) is equal to the sum of the flows in the afferent arterioles (*C*); without autoregulation, it is 6.92 mL·min^{-1}, and with autoregulation, it is 6.87 mL·min^{-1}.

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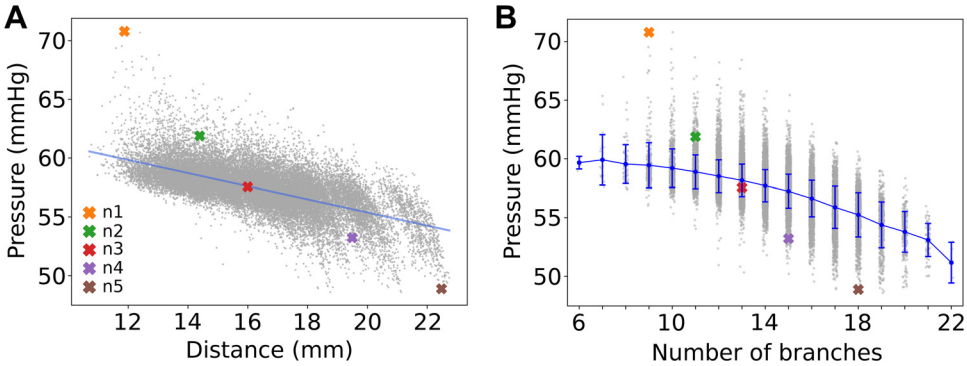


Figure 8. Scatter plots of glomerular capillary pressures of the 29,608 nephrons with autoregulation. *A*: glomerular capillary pressure vs. accumulated length along the vasculature from the root. Blue line is the best linear fit. *B*: glomerular capillary pressure vs. number of vascular branches from the root. Blue line connects the mean values of each group, and the vertical lines are the standard deviations for each group. The colored crosses labeled n1–n5 represent the selected nephrons in Figs. 9 and 12 with glomerular capillary pressures from high (n1) to low (n5).

the relative change of renal blood flow (RBF) change and the relative change of renal arterial pressure (P_{art}):

$$\mathrm{ARI} = \frac{(\mathrm{RBF}_2 - \mathrm{RBF}_1)/\mathrm{RBF}_1}{(\mathrm{P}_{art2} - \mathrm{P}_{art1})/\mathrm{P}_{art1}}. \qquad (20)$$

An ARI of zero implies perfect autoregulation, as there is no change in blood flow when the arterial pressure is increased. The higher the value of ARI, the less efficient the autoregulation, with a value of ARI ≥ 1 indicating no autoregulation. Figure 12 shows ARI in the nephron-vascular network at increasing levels of arterial pressure. As can be seen from Fig. 12, the model predicts an ARI of $\approx$0.2 in the arterial pressure range from 100 mmHg to 160 mmHg.

Single nephron model.

For comparison with the full-scale renal arterial tree model, Fig. 12 shows autoregulatory curves and corresponding ARI's for the single nephron model corresponding to the three different nephrons as discussed in *Autoregulation in Single Nephron Model*. The three nephrons, n2, n3, and n4, correspond to the middle 3 of the 5 nephrons shown in Fig. 9. It is clear that the autoregulatory behavior of the single nephron model depends on which nephron is chosen from the full-scale renal arterial tree. If the single nephron is chosen with a medium pressure initially (n3, red curve), the autoregulatory efficiency is high. On the contrary, if the single nephron is chosen with a high glomerular capillary pressure initially (n2, green curve), then autoregulation is poor when the arterial pressure becomes high, even reaching an ARI > 1. Finally, if the single nephron initially has a low glomerular capillary pressure (n4, purple curve), autoregulation is moderate at high arterial pressures, but poor at low arterial pressures.

Nephron subpopulations.

Figure 12 shows renal blood flow autoregulation after integrating three subpopulations of nephrons, superficial, midcortical, and juxtamedullary. It is clear that using a more realistic scenario for the nephrons yields a slight improvement in autoregulatory efficiency, especially at the highest arterial pressures.

To take a closer look at the effect of having three nephron populations, Fig. 13 shows the behavior of the three types of nephrons at an arterial pressure of $P_{art} = 140$ mmHg. From Fig. 13 (*top*), it is clear that the NaCl concentrations at the macula densa (C_S(md)) are similar in all three types when the flows are around the control case (around 4 nL·min^{-1}). However, when flow increases, juxtamedullary nephrons with long Henle loops will have higher NaCl concentrations at the macula densa (C_S(md)). Figure 13 (*bottom*) shows the TGF response versus NaCl concentration levels at the macula

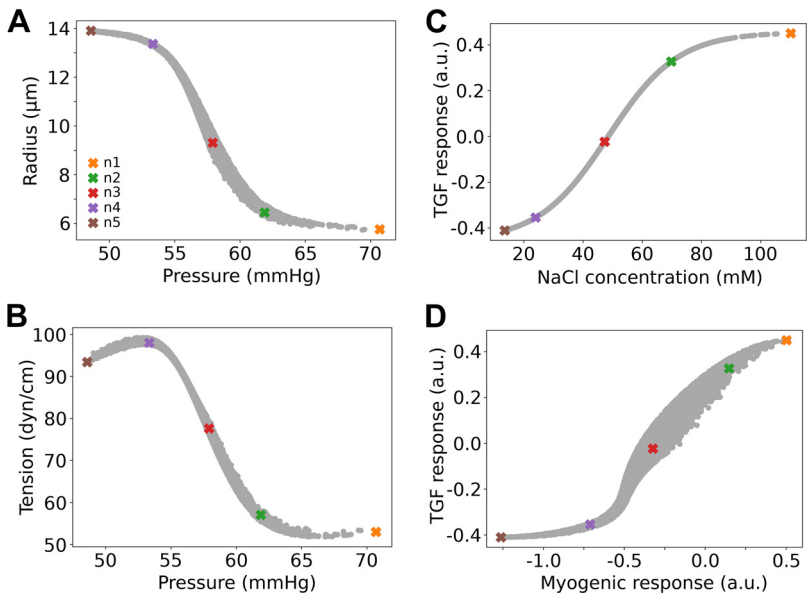


Figure 9. Scatter plots in the control case with an arterial pressure of $P_{art} = 100$ mmHg (all nephrons are of superficial type) with activated autoregulation: glomerular capillary pressure vs. radius (*A*), tension vs. glomerular capillary pressure (*B*), TGF response vs. NaCl concentration at macula densa [C_S(md); C], and TGF response vs. myogenic response (*D*). The colored crosses in the four panels indicate five different nephrons operating under different glomerular capillary pressures [from high (n1) to low (n5)] in the vascular tree.

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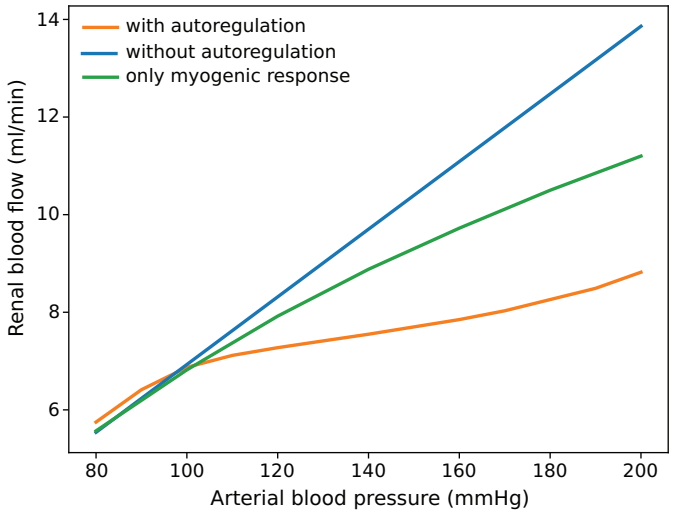


Figure 10. Renal blood flow as a function of arterial blood pressure increases linearly with pressure when there is no autoregulation (blue). When autoregulation is active, the blood flow reaches a plateau and stays relatively constant over a wide range of arterial pressures (orange). If only the myogenic mechanism is active, the renal blood flow is intermediate between the two previous cases (green).

densa $C_S(\mathrm{md})$ in the three types of nephrons. They all saturate when $C_S(\mathrm{md})$ becomes too large, but the juxtamedullary nephrons with long loops of Henle saturate at a higher value for the TGF response.

Renal artery stenosis.

In addition to simulating the ability of autoregulation to stabilize blood flow under different input pressure levels, another perspective in full-scale modeling of the nephron-vascular network is to simulate pathological conditions in the vasculature. In renal arteries, a common pathology is renal artery stenosis (RAS), which is a narrowing of the lumen of the renal artery, most often caused by atherosclerosis or fibromuscular dysplasia (40). RAS is a serious condition that can cause hypertension and renal damage due to ischemia. For example, the American Heart Association defines a RAS of more than 60% to be a critical degree of stenosis (41).

One of the main purposes of simulating RAS is to examine the consequences of different degrees of RAS and the ability of the kidney to compensate for the stenosis. By simulating different levels of stenosis and studying the corresponding outcome on renal hemodynamics, better preventive and surgical strategies can be made (40, 42, 43).

Initially, RAS is simulated in the full-scale model of the nephron-vascular network using the control case arterial pressure ($P_{art} = 100$ mmHg) while gradually decreasing the diameter of renal artery (the root vessel in the full-scale tree model).

As shown in Fig. 14 (*top*), renal blood flow remains relatively constant at the lower degrees of RAS, but drops dramatically when stenosis exceeds 50%, regardless of whether or not autoregulation is active.

Since RAS often occurs in conjunction with hypertension, the consequences of RAS were also simulated using an arterial pressure of $P_{art} = 140$ mmHg. Figure 14 (*bottom*) shows that in hypertension, the autoregulatory mechanism is effective in stabilizing the renal blood flow when the stenosis is $<70\%$. Without autoregulation renal blood flow already decreases at a stenosis level of around 50%. However, it can also be seen that with a high level of stenosis, even at a pressure level where the autoregulatory mechanisms are highly effective, the impact of autoregulation on renal blood flow remains limited.

DISCUSSION

In this work, we tested the usability of a reconstructed renal vascular tree in simulating the renal response to varying arterial pressures or varying degrees of renal artery stenosis. The full-scale renal arterial tree was reconstructed using a constructive physiologically based optimization method that incorporates anatomical data extracted from a scan of a rat kidney (18). The model of the vascular tree was then combined with 29,608 nephron models to allow a simulation of renal hemodynamics on a scale that had not been achieved before.

A major problem when trying to construct a model of the renal vascular tree is the lack of information on the dimensions of the vascular segments and the geometry of the vascular tree itself. Currently, the only method to obtain a complete picture of the renal vasculature is the use of casting methods, where the experimental method directly affects the vessels and therefore their dimensions, especially their radii (19). Two previous papers have attempted to reconstruct a full-scale renal vascular tree (44, 45). Using data from Ref. 19, Kleinstreuer et al. (45) conducted a modeling study of myogenic autoregulation in a full-scale model of the renal vascular tree. However, they did not recreate the branching structure of the renal vascular tree but simply assumed that the vessels of Strahler order $i-1$ originate from the vessels of Strahler order i. As can be seen in Fig. 3, this is not true in all cases. For example, a vessel of Strahler order 0 may branch from a vessel of Strahler order 2. In contrast, Deng and Tsubota (44) reconstructed the renal vascular tree using 2 types of branch models, one for large arteries and one for small arteries. Consequently, the resulting vascular tree is deterministic and has a high degree of symmetry that is not seen in a real renal vascular tree. In contrast, the approach (18) used in the present paper leads to an asymmetric vascular tree, where the number of branches between the root and the terminal nodes varies, as seen in a real renal vascular tree (3, 19). The present work also extends the previous studies by including both TGF and the myogenic response.

In the present model, TGF acts by modifying the setpoint of the myogenic response (see *Eq. 11*). This differs from previous models (6, 30) where the two mechanisms were assumed to be additive. The new formulation is motivated by the fact that the two mechanisms converge on the vascular smooth muscle cell, and both influence the contractile machinery through changes caused by changes in the membrane potential (46, 47). Therefore, the responses are not simply independent and additive, as has also been shown experimentally (48). Furthermore, the new formulation avoids the antagonism between the mechanisms when a simple additive model is used (30).

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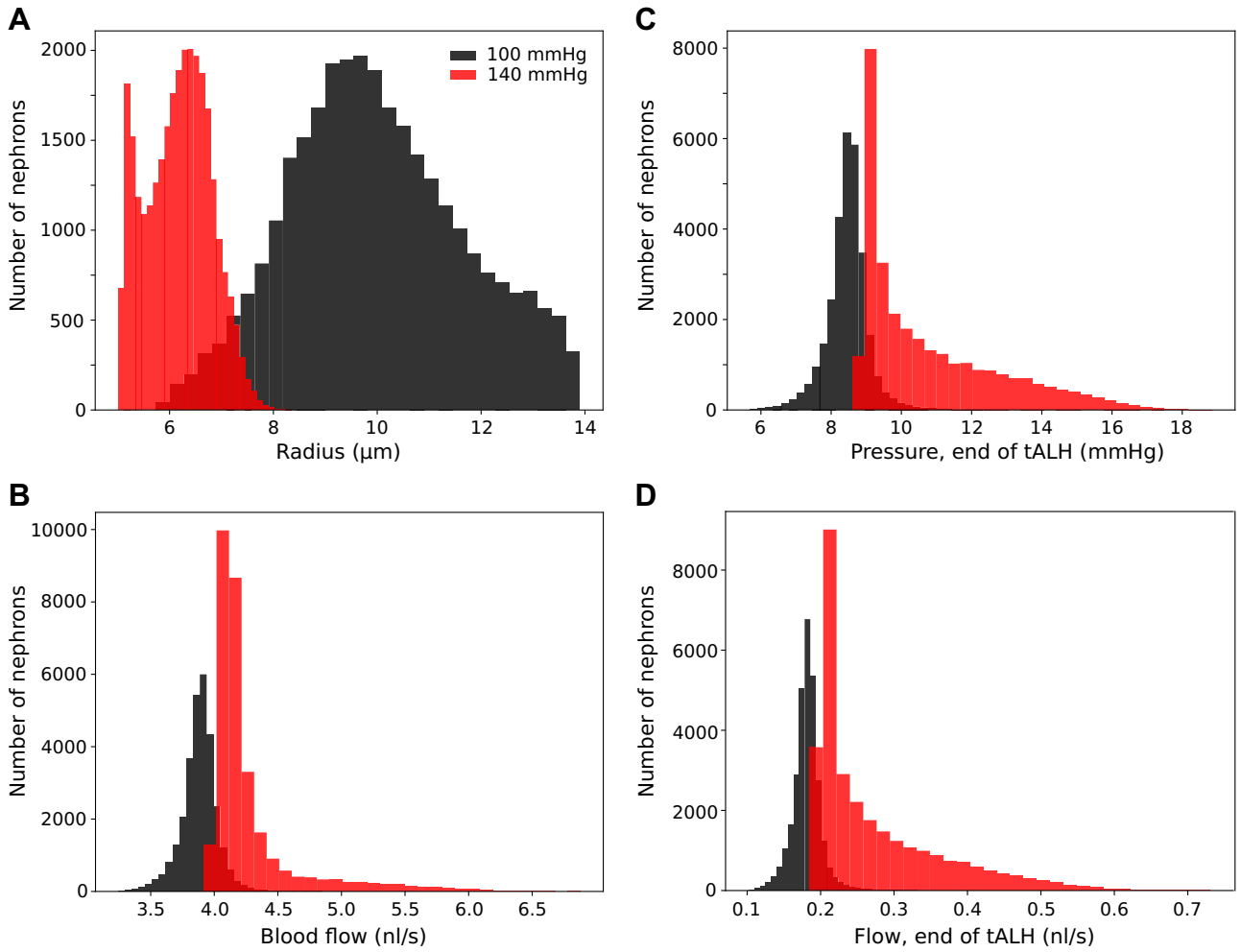


Figure 11. Histograms showing the effect of autoregulation on the distributions of afferent arteriolar radii (*A*), afferent arteriolar blood flow (*B*), tubular pressure (*C*), and tubular flow rate (*D*) at the end of the ascending limb of the loop of Henle at renal arterial pressures of 100 mmHg (black) and 140 mmHg (red). Total renal blood flow (RBF) is equal to the sum of the flows in afferent arterioles (*B*); at 100 mmHg, it is 6.87 mL·min^{-1}, and at 140 mmHg, it is 7.55 mL·min^{-1}.

As can be seen in Fig. 7*A*, the distribution of pressures at the ends of the afferent arterioles without autoregulation is very broad, indicating that the reconstructed vascular tree by itself produces an unrealistic pressure distribution. However, when nephrons, together with their intrinsic regulatory mechanisms, are added to the vascular tree, the result is a much narrower and more realistic distribution of terminal pressures (Fig. 7*A*). So, although we lack detailed information on the dimensions of the smallest vessels in the rat kidney, by combining the vascular tree model with the regulating nephron models, we were able to obtain what appears to be a reasonable pressure distribution.

Since in the present model nephrons have identical parameter values for myogenic and TGF responses (see Tables 3 and 5), the final variation in their state, as shown in Figs. 7 and 9, is due to the position of the glomeruli within the vascular tree. As shown in Fig. 8, nephrons closer to the root will generally have a higher glomerular capillary pressure for structural reasons than nephrons farther away, and this difference will only be partially affected by the autoregulatory mechanisms. As a consequence, their operating points will differ, with some of the nephrons at the extreme of their responsiveness already in the control situation, affecting the ability to autoregulate blood flow (see below).

Our simulation results in Figs. 10 and 12 show strong autoregulation due to the joint feedback from TGF and the myogenic response. In general, autoregulation is most effective in counteracting an increase in arterial blood pressure from 100 mmHg to 160 mmHg, whereas there is little autoregulation when pressure is reduced (< 100 mmHg). The range of autoregulation is limited due to the saturation of especially TGF, and autoregulation saturates when the pressure becomes too high (> 160 mmHg) or too low (< 100 mmHg).

Many textbooks depict perfect renal blood flow autoregulation (49, 50), that is, that blood flow stays constant within the pressure range in Fig. 12. However, in most experiments, a well-functioning autoregulation usually has an ARI around 0.2, whereas an ARI < 0.2 indicates excellent autoregulation (51–53), which matches our simulation results, where the ARI is close to 0.2 with an arterial pressure ranging from 110 to 160 mmHg.

In the present model, we assumed that the absolute proximal reabsorption rate stayed constant during changes in perfusion pressure. As a consequence, the fractional proximal

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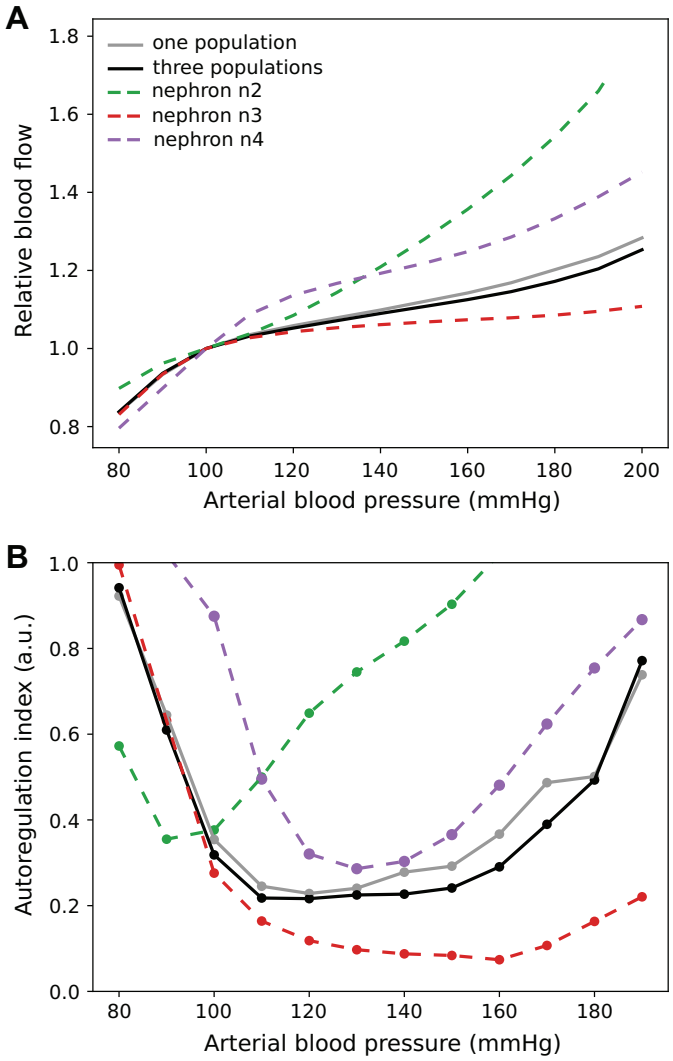


Figure 12. Renal autoregulation for one and three populations of nephrons in the full-scale vascular network (solid curves) as well as individual nephrons in a single nephron model (dashed curves). The color of the curves for individual nephrons is the same as in Fig. 9. *A*: renal blood flow. For comparison, the blood flow values are normalized to the value at 100 mmHg. *B*: autoregulation index, calculated over pressure intervals of 10 mmHg, at various levels of arterial blood pressure. Although individual nephrons show diverse autoregulatory properties, the full-scale vascular tree model with three populations of nephrons shows a wider plateau within a physiologically relevant range of pressures.

reabsorption rate decreases as perfusion pressure increases, due to the slight increase in the glomerular filtration rate. Chou and Marsh (54) showed that increasing the arterial pressure from 105 mmHg to 125 mmHg was associated with a 50% increase in outflow from the proximal tubule, indicating a decrease in proximal fractional reabsorption. In another study, Chou and Marsh (55) showed that an increase of arterial pressure of only 24 mmHg reduced proximal reabsorption by 44%. A similar conclusion was reached in a rat micropuncture study by Bank et al. (56). In addition, McDonough and Nguyen (for a review, see Ref. 57) have in an impressive series of studies shown that acute hypertension leads to a redistribution of several ion transporters in the proximal tubule leading to a reduced rate of proximal reabsorption and that this is a central mechanism behind pressure natriuresis. By assuming a constant value for the proximal reabsorption rate, we obtain a conservative estimate of autoregulatory efficiency. If one assumes not only a decrease in the fractional proximal reabsorption rate as in the present model but also a decrease in the absolute rate of proximal reabsorption, it is possible to obtain superautoregulation (ARI < 0) (58).

According to Guyton (59), pressure natriuresis and diuresis are essential for long-term control of the arterial pressure. Since the present model does not include the most distal parts of the tubular system, the effect of increasing the renal perfusion pressure on NaCl and water excretion cannot be assessed directly. However, as shown in Fig. 11, increasing the perfusion pressure from 100 mmHg to 140 mmHg results in a 33% increase in median tubular flow rate at the end of the thick ascending limb, indicating an increased load of NaCl and water to the collecting ducts in accordance with experimental results (54, 55, 60). The pressure and flow at the end of the thick ascending limb were determined by a nonlinear boundary condition that represents the flow resistance in the more distal tubular segments as that of a single compliant tubule;

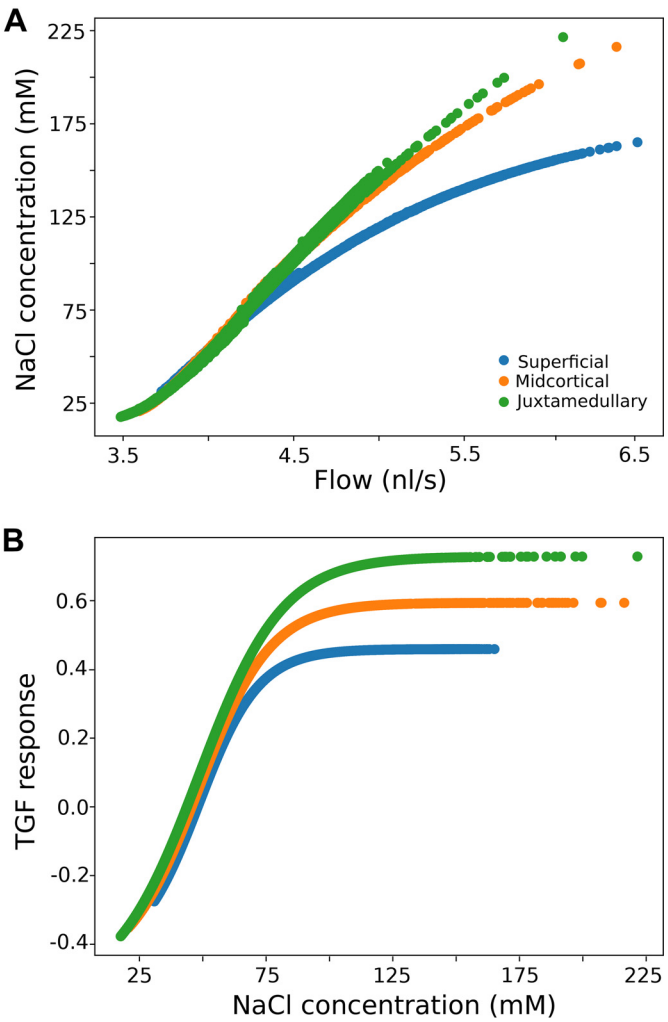


Figure 13. Scatter plots showing the relations between the NaCl concentration at the macula densa [C_S(md)] vs. tubular flow rate (*A*), and the TGF response vs. C_S(md) for the three subpopulations of nephrons (*B*), $P_{art} = 140$ mmHg.

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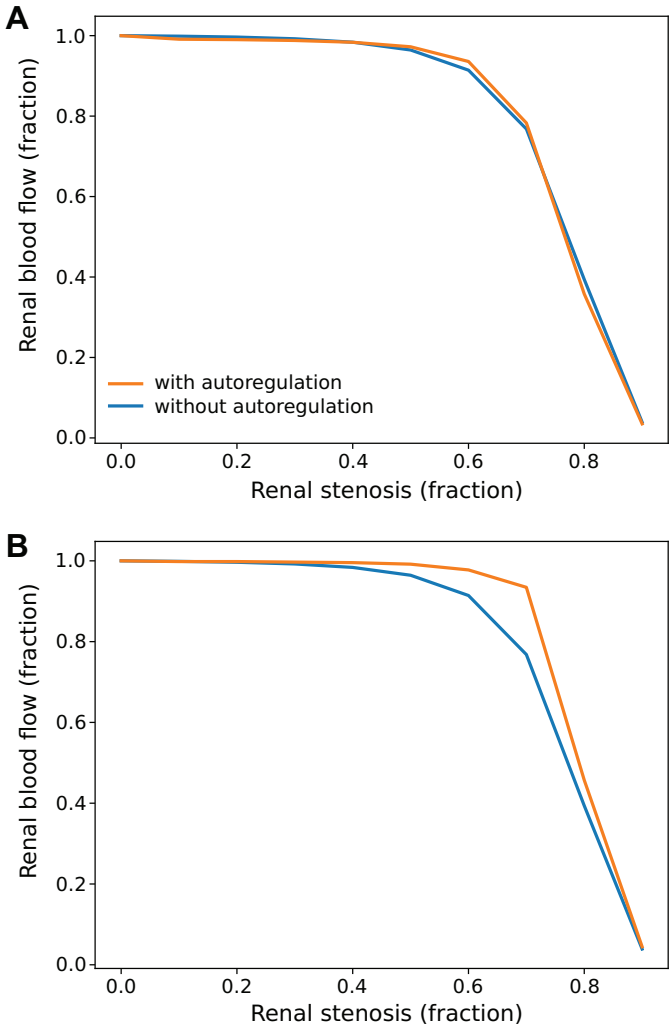


Figure 14. Renal blood flow at various stenosis levels in the control case (*A*) with an arterial pressure $P_{art} = 100$ mmHg and in the hypertensive case (*B*) with an arterial pressure $P_{art} = 140$ mmHg. The control case shows an abrupt decrease of renal blood flow with around 50% stenosis whether or not autoregulation is active. In the hypertensive case autoregulation postpones the abrupt decrease of blood flow from around 50% stenosis to around 70% stenosis.

see *Eq. A21*. It is based on empirical data (59), where the flow in all nephrons was changed in parallel (volume loading or osmotic diuresis) and incorporates the effects of a simultaneous increase in flow in all collecting ducts.

To compare our full-scale model to that of a single nephron model, we performed a set of simulations of renal autoregulation using only the single nephron model. As can be seen in Fig. 12, the single nephron model produces perfect autoregulation, but only when its parameters are chosen so that the TGF and myogenic mechanisms have optimal working conditions (corresponding to nephron n3). This is expected since in the full-scale model, the nephrons operate under different combinations of flow and pressure depending on their location within the renal vascular tree. As can be seen in Fig. 12 if the conditions are slightly off with respect to the operating conditions of the nephron (nephrons n2 and n4), the pressure range over which autoregulation is effective is both shifted and reduced. Therefore, it is no surprise that a full-scale model with its heterogeneity among the nephrons will be less effective than a single nephron model when it comes to autoregulatory efficiency. Consequently, the use of a single nephron model can lead to an overestimation of autoregulatory efficiency.

The relative roles of TGF and the myogenic response in renal autoregulation have been the subject of many studies over the last decades (53). Today, the consensus is that they both contribute to autoregulation (53). Experimentally, it is not possible to block the myogenic mechanism, since prevention of the myogenic response, for example, by blocking voltage-operated calcium channels, abolishes vascular responses to both autoregulatory mechanisms. Studies in which TGF has been blocked by, for example, the administration of a loop diuretic show a contribution of around 25%–35% (53). This is in good agreement with the present model. When simulating autoregulation without the TGF response (Fig. 12), autoregulation is significantly impaired. This shows that both mechanisms are necessary for efficient autoregulation of renal blood flow.

The nephron population is not homogeneous but shows considerable variation, especially with regard to the length of the loop of Henle (Fig. 5). We have incorporated this into the model by having three subpopulations of nephrons. Most nephrons have short loops that extend only to the outer medulla, and their glomeruli reside predominantly in the upper part of the cortex. The second group has loops of Henle of intermediate length, with the bend in the inner medulla. Finally, long-looped nephrons have a loop of Henle that reaches all the way to the tip of the papilla. The inclusion of the three subpopulations was associated with an improvement in autoregulatory efficiency (see Fig. 12). This effect is primarily due to the larger operating range in the nephrons with longer loops (33, 34). Thus, the improvement in renal autoregulation seen when having three subpopulations of nephrons is due to the slightly stronger TGF response in the midcortical and juxtamedullary nephrons compared with the superficial ones (see Fig. 13). However, as Fig. 12 shows, nephrons with longer loops of Henle only have a moderate effect on the overall efficiency of renal blood flow autoregulation. This is to be expected since the TGF response is only moderately stronger in the midcortical and the juxtamedullary nephrons compared with the superficial nephrons.

In the present model, for simplicity, each nephron within the same subpopulation has the same length for the loop of Henle. In the real kidney, the length of the loops of Henle varies within each subpopulation. Since the effect on autoregulation of having three subpopulations of nephrons compared with just one was modest, we do not anticipate that adding variable lengths for the loops of Henle within each subpopulation will significantly affect the autoregulatory efficiency of the model.

Another simplification is the use of only a single solute, NaCl, in the tubular fluid. The omission of urea will lead to a slight overestimation of the concentration of NaCl at the end of the descending limb. However, the amount of urea in the tubular fluid at the end of the descending limb is quite low ($\sim$40 mosM·L^{-1}) compared with NaCl ($\sim$1,160 mosM·L^{-1}), so the error is $<$4% and will only have a small effect on the NaCl load from the descending to the ascending limb. The

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NaCl reabsorption in the ascending limb is load dependent, and the error will be reduced further as the fluid moves up the ascending limb toward the macula densa. Furthermore, the reabsorption of NaCl in the ascending limb of the loop of Henle, and thereby the signal to the macula densa, is not influenced by the concentration of urea in the tubular fluid. It therefore seems reasonable to ignore the presence of urea in the tubular fluid for the purpose of the present model. The significance of urea for the signal to the macula densa lies in its presence in the inner medullary interstitium, where it is responsible for ~50% of the osmolarity. Because of the much higher levels of urea in the inner medullary interstitium compared with the corresponding tubular fluid, there is an outward gradient for NaCl that drives passive reabsorption of NaCl in the thin ascending limbs of the loop of Henle. This aspect of urea is included in the model through the assumptions that the NaCl concentration is constant in the inner medullary interstitium and that the increase in osmolarity toward the tip of the papilla is due to an increasing concentration of urea, see Fig. A1.

Computer simulation of the full-scale nephron-vascular network can also be used to assess the impact of different physiological and pathological conditions in the kidney and help researchers and clinicians investigate various aspects of blood flow disruptions, for example, the consequences of thrombosis or atherosclerosis in the renal vasculature. As an example, we have simulated renal artery stenosis to examine its effect on renal blood flow. Therefore, we also tested the usefulness of the full-scale model in simulating pathological conditions in the renal vasculature, specifically renal artery stenosis.

Several efforts have been made to simulate renal artery stenosis (RAS) using computational fluid dynamics (CFD) (40, 42, 43, 61, 62), which is the conventional method used to simulate blood flow and blood flow dysfunction in the vasculature. These methods involve constructing computational meshes from images of the larger renal vessels. This is then used to numerically solve the Navier–Stokes equations for the system under investigation. This approach has been used extensively in studies of coronary artery disease and aortic aneurysms (63–67). The difference between the approach in this paper and the previous work is that, although the latter has focused only on the effect of stenosis in the renal artery (40, 42, 61), the present approach simulates blood flow in a full-scale model of the renal vasculature together with the impact of renal autoregulation. Since RAS usually only affects the diameter of the renal artery, it is possible to use CFD-based methods to simulate renal artery stenosis with only one segment of the renal arterial network. However, our full-scale nephron-vascular network also accounts for the effects of autoregulation. Note that although RAS involves the decrease in diameter of only the renal artery, we need a full-scale structure to properly define the outlet boundary conditions and solve for the resulting renal blood flow.

Our simulation results show a significant decrease in renal blood flow with stenosis of more than 50%, which is in good agreement with CFD-based simulations (40, 42, 43, 61, 62). This could indicate that complex computational meshes or finite-element or volume solvers are not necessary, as a tree-based model can generate very similar results and is significantly more efficient.

The impact of autoregulation in preventing a drop in blood flow is minimal (see Fig. 14, *top*). This is expected because in the control case (P_{art} = 100 mmHg), autoregulation has limited power to compensate for a further decrease in renal perfusion pressure (see Fig. 12). Consequently, as the stenosis becomes severe, renal blood flow will be reduced. Therefore, the flat curve in Fig. 14, when the stenosis < 50%, is not mainly attributable to autoregulation but rather stems from the fact that the hemodynamic resistance of the renal artery has little impact on renal blood flow until the stenosis is severe. At higher arterial pressures, where autoregulatory mechanisms are more effective, an effect of autoregulation in maintaining renal blood flow is detectable, although the impact on overall renal blood flow remains limited.

The present model has several limitations in that it does not include the dynamics of the autoregulatory mechanisms and interactions between nephrons. The current model is a steady-state model. It is well known that, at the microcirculatory level, renal hemodynamics show significant dynamics, such as oscillations in single nephron blood flow and glomerular filtration rate (1). The aim of the present model was to test the suitability of the generated vascular tree for modeling the well-known steady-state autoregulatory response of the whole kidney. This is independent of the dynamics of the individual autoregulatory mechanisms. The inclusion of dynamic elements in the model is a logical next step for future work, as it will allow a study of the fascinating phenomena of nephron-nephron synchronization that has been observed in the rat kidney (68–70). This will require solving ordinary and partial differential equations for all nephrons within the full-scale vascular model, a computationally intensive task.

Although our model encompasses all afferent arterioles and nephrons within a complete tree structure, we have yet to address possible interactions between nephrons. Our model includes an interdependence of the afferent arteriolar responses because of the hemodynamic coupling of nephrons within the arterial tree. However, it ignores interactions arising from spread of electrical signals along vessel walls (13, 15). The cells of the afferent arterioles are electrically coupled due to the presence of gap junctions between the cells. This allows for the spread of the TGF signal from one nephron into neighboring nephrons deriving their afferent arterioles from the same terminal arteriole (47). As a result, the TGF response of the ensemble exceeds that of the individual nephrons (71).

A second source for possible interactions between nephrons not included in the model is the convergence of nephrons in the distal part of the tubular system. Starting at the connecting tubules, nephrons begin to merge, and the tubular fluid flows coalesce, through multiple intermediary steps, into a few inner medullary collecting ducts that end on the tip of the papilla. If one nephron changes its tubular flow rate, then the flow, and thereby the pressure, will change in the corresponding connecting tubule and collecting duct. This will affect the other nephrons emptying into the same region, thus causing changes in their tubular pressures and flows. In a multi-nephron model (36,000 nephrons), this merger was explicitly modeled to account for a possible interaction between nephrons of different classes and it was found that it could lead to a weak interaction between nephrons (72).

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Finally, changes in tubular reabsorption rates in a group of nephrons may also lead to changes in the composition and pressure of the renal interstitium, including the medullary interstitium, which ultimately could affect tubular reabsorption of NaCl and water in a large group of neighboring nephrons. A full model of the kidney will have to include all of the above sources of nephron-nephron interactions.

To summarize, the full-scale model integrating a realistic vascular network topology of the kidney and its hemodynamic regulatory mechanisms has shown: *1*) nephrons operate under a wide range of pressures due to the structure of the vascular tree with perfusion pressures decreasing toward the kidney surface; *2*) the myogenic mechanism and the TGF response both contribute to autoregulation; *3*) the autoregulatory mechanisms adjust the resistances of the afferent arterioles to reduce the differences in operating conditions between nephrons in the vascular tree; *4*) the contribution of individual nephrons to autoregulation of renal blood flow depends on their operating conditions; *5*) long-looped nephrons improve autoregulation by extending its pressure range; *6*) the use of a single nephron model can lead to an overestimation of autoregulatory efficiency; *7*) stenosis of the renal artery has little effect on renal blood flow until it reaches a critical level of ~50% reduction in vessel radius; and *8*) autoregulation has no effect or only a small effect in modulating the response to renal artery stenosis.

In conclusion, the present work has demonstrated the usability of a reconstructed renal vascular tree in the construction of a full-scale model of the regulation of renal hemodynamics. The model gives a realistic picture of both renal autoregulation of blood flow and the renal hemodynamic response to renal artery stenosis. This is a first step toward the development of "the virtual kidney."

APPENDIX A: MODEL DETAILS

Proximal Tubule

In the proximal tubule, the concentration of NaCl, $C_S(z)$, remains constant due to isosmotic transport where the reabsorption rates of water and NaCl are proportional, thus $C_S(z) = C_S(0)$ throughout the proximal tubule. The volume reabsorption, J_V, is described by an exponentially decreasing function along the length of the tubule, where κ and θ are empirical constants:

$$J_V(z) = \kappa \exp(-\theta z) \tag{A1}$$

with 0 cm $\leq z \leq$ 1 cm corresponding to the start and the end of the proximal tubule, respectively. Thus,

$$\frac{dQ_T}{dz} = -\kappa \exp(-\theta z)\,. \tag{A2}$$

The flow $Q_T(z)$ at a given point in the proximal tubule can be found by solving the integral

$$Q_T(z) = \int_0^z \frac{dQ_T}{dz}dz = \int_0^z -\kappa \exp(-\theta z)dz = \frac{\kappa}{\theta}\exp(-\theta z) + C. \tag{A3}$$

Given the boundary condition $Q_T(z = 0)$, we can find C, giving

$$\frac{\kappa}{\theta} + C = Q_T(0) \Rightarrow C = Q_T(0) - \frac{\kappa}{\theta}\,. \tag{A4}$$

Substituting the expression for C into *Eq. A3* gives the flow in the proximal tubule:

$$Q_T(z) = \frac{\kappa}{\theta}\exp(-\theta z) + Q_T(0) - \frac{\kappa}{\theta}\,. \tag{A5}$$

Likewise, substituting into *Eq. 4* gives the equation for the tubular pressure:

$$\frac{dP_T}{dz} = -\frac{8\mu}{\pi r^4}\left(\frac{\kappa}{\theta}\exp(-\theta z) + Q_T(0) - \frac{\kappa}{\theta}\right). \tag{A6}$$

This can be solved analytically:

$$P_T(z) = \int_0^z \frac{dP_T}{dz}dz = \frac{8\mu\kappa}{\pi r^4\theta^2}\exp(-\theta z) - \frac{8\mu}{\pi r^4}\left(Q_T(0) - \frac{\kappa}{\theta}\right)z + K. \tag{A7}$$

Here, however, the boundary condition for the tubular pressure is given at the distal end ($Z = 1$ cm) with the value $P_T(Z)$. Thus,

$$P_T(Z) = \frac{8\mu\kappa}{\pi r^4\theta^2}\exp(-\theta Z) - \frac{8\mu}{\pi r^4}\left(Q_T(0) - \frac{\kappa}{\theta}\right)Z + K \tag{A8}$$

giving

$$K = P_T(Z) - \frac{8\mu\kappa}{\pi r^4\theta^2}\exp(-\theta Z) + \frac{8\mu}{\pi r^4}\left(Q_T(0) - \frac{\kappa}{\theta}\right)Z. \tag{A9}$$

Substituting into *Eq. A10* gives the solution for pressure in the proximal tubule:

$$P_T(z) = \frac{8\mu\kappa}{\pi r^4\theta^2}\left(\exp(-\theta z) - \exp(-\theta Z)\right) + \frac{8\mu}{\pi r^4}\left(Q_T(0) - \frac{\kappa}{\theta}\right)(Z - z) + P_T(Z)\,. \tag{A10}$$

Given an inflow to the tubule ($Q_T(0)$), the flow in the proximal tubule is calculated using *Eq. A5*. Using the boundary condition $P_T(Z)$, the pressure in the proximal tubule is calculated from *Eq. A10*.

NaCl Concentrations and Osmolarities in the Medulla

Figure A1 shows an overview of the model for the renal medulla. The thin descending limb starts at the border between the outer and inner stripe of the outer medulla. For superficial nephrons, the loop bends at the border between the outer and inner medulla, whereas for midcortical nephrons, the bend is midway down the inner medulla. Finally, for juxtamedullary nephrons, the bend is in the deepest part of the inner medulla, corresponding to the tip of the renal papilla. The interstitial fluid in the cortex and the outer stripe of the outer medulla has an osmolarity of 300 mosM·L^{-1}, corresponding to a NaCl concentration of 150 mM (see Fig. A1) (73). In the inner stripe of the outer medulla, the osmolarity and the NaCl concentration increase in parallel, reaching values of 600 mosM·L^{-1} and 300 mM for the osmolarity and the NaCl concentration at the border between the outer and inner medulla (see Fig. A1). In the inner medulla, there is a further increase in osmolarity to 1,200 mosM·L^{-1} in its deepest part. This is due to an increase

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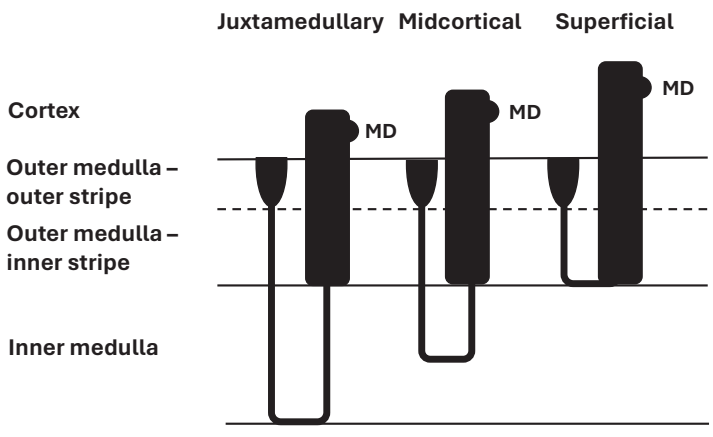


Figure A1. Model for the renal medulla. The tubular segments found within the medulla are (in order of the flow): the last part of the proximal tubule, the thin descending limb, the thin ascending limb (only in midcortical and juxtamedullary nephrons), and the thick ascending limb. The macula densa (MD) is localized close to the end of the superficial part of the thick ascending limb. C_I^{osm}, the osmolarity in the interstitial fluid; C_I^{NaCl}, the concentration of NaCl in the interstitial fluid. The figure is not drawn to scale; see Table 6. See text for details.

in the concentration of urea, whereas the NaCl concentration remains at 300 mM throughout the inner medulla (73).

Descending Limb of the Loop of Henle

In the descending limb of the loop of Henle, water is reabsorbed by osmosis, whereas there is diffusion of NaCl from the interstitium (I) into the tubule:

$$J_V(z) = L_v(C_I^{osm}(z) - n_s C_S(z)) \tag{A11}$$

and

$$J_S(z) = L_S(C_S(z) - C_I^{NaCl}(z)) \tag{A12}$$

where $C_I^{osm}(z)$ is the osmolarity and $C_I^{NaCl}(z)$ is the NaCl concentration in the medullary interstitium, see Fig. A1. C_S is the concentration of NaCl in the tubular fluid and $n_s = 2$ corresponding to two osmotic particles per NaCl molecule. Substituting into *Eq. 4* gives a system of three coupled ordinary differential equations for pressure, flow, and NaCl concentration

$$\frac{dP_T}{dz} = -\frac{8\mu}{\pi r^4} Q_T$$

$$\frac{dQ_T}{dz} = -L_v(C_I^{osm} - n_s C_S)$$

$$Q_T \frac{dC_s}{dz} + C_s \frac{dQ_T}{dz} = -L_S(C_S - C_I^{NaCl})$$

which, after rearranging the third equation, gives:

$$\frac{dP_T}{dz} = -\frac{8\mu}{\pi r^4} Q_T \tag{A13}$$

$$\frac{dQ_T}{dz} = -L_V n_s(C_I^{osm} - C_S) \tag{A14}$$

$$\frac{dC_S}{dz} = -\frac{L_S\left(C_S - C_I^{NaCl}\right) + C_S \frac{dQ_T}{dz}}{Q_T} \tag{A15}$$

with the boundary values $Q_T(0)$ and $P_T(Z)$ for the flow at the start and the pressure at the end ($z = Z$) of the descending limb of the loop of Henle, respectively. Note that the inflow to the descending limb is equal to the outflow from the proximal tubule, see above.

Ascending Limb of the Loop of Henle

In the ascending limb, only NaCl is reabsorbed, so the NaCl concentration C_S decreases, whereas the flow Q_T remains constant and equals the flow at the end of the descending limb $Q_T(\text{desc_end})$. Here

$$J_S = L_S(C_S - C_I^{NaCl}) + \frac{V_{max} C_S}{K_m + C_S} \tag{A16}$$

where the first term represents passive diffusion driven by the concentration gradient between the tubular fluid and the interstitium, and the second term represents active transport across the tubule. In the thin ascending limb, there is only passive reabsorption of NaCl, that is, $V_{max} = 0$, whereas in the thick part, there is passive and active reabsorption of NaCl.

Substituting *Eq. A16* into *Eq. 4* gives a differential equation for C_S:

$$Q_T \frac{dC_s}{dz} + C_s \frac{dQ_T}{dz} = -L_S(C_S - C_I^{NaCl}) - \frac{V_{max} C_S}{K_m + C_S}. \tag{A17}$$

Since $Q_T = Q_T(\text{desc_end})$ stays constant, $\frac{dQ_T}{dz} = 0$, the above differential equation can be simplified to

$$\frac{dC_s}{dz} = \frac{-L_S(C_S - C_I^{NaCl}) - \frac{V_{max} C_S}{K_m + C_S}}{Q_T(\text{desc_end})}. \tag{A18}$$

Similarly, the pressure can be simplified to:

$$\frac{dP_T}{dz} = -\frac{8\mu}{\pi r^4} Q_T(\text{desc_end}) \tag{A19}$$

which can be solved analytically by

$$P_T(z) = \int_0^z \frac{dP_T}{dz'} dz' = -\frac{8\mu}{\pi r^4} Q_T(\text{desc_end}) z + C. \tag{A20}$$

The boundary condition for the previous equation is the tubular pressure at the end of the loop of Henle ($z = Z$). The value of Z depends on whether the nephron has a short, intermediate, or long loop of Henle, see Table 6. The pressure, $P_T(Z)$, is given by the empirical relationship (27)

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$$P_T(Z) = \frac{Q_T(Z)}{(\alpha P_T(Z) + \beta)^4} \tag{A21}$$

where $Q_T(Z) = Q_T(\text{desc_end})$, since flow is constant in the ascending limb, and α and β are empirical constants. Thus,

$$C = P_T(Z) + \frac{8\mu}{\pi r^4} Q_T(\text{desc_end}) Z. \tag{A22}$$

Substituting this into *Eq. A20* gives an analytical solution for pressure in the ascending limb of the loop of Henle:

$$P_T(z) = \frac{8\mu}{\pi r^4} Q_T(\text{desc_end})(Z - z) + P_T(Z). \tag{A23}$$

NaCl Concentration at the Macula Densa

One of the main purposes of modeling the NaCl concentration (C_S) in tubular fluid is that the activity of tubuloglomerular feedback depends on the NaCl concentration at the macula densa $C_S(\text{md})$, see *Eq. 12*. In our model, the macula densa is located at $z = 0.5$, $z = 0.65$, or $z = 0.82$ in the thick ascending limb, depending upon whether the nephron has a short, intermediate, or long loop of Henle, see Table 6.

Wall Thickness of the Afferent Arteriole

The wall thickness, $h(r_v)$ in the elastic tension term T_e, varies with the radius r_v so that the cross-sectional area of the vessel wall remains constant. Specifically, given reference values for radius, r_0 and wall thickness, h_0, $h(r_v)$ is given by:

$$\begin{aligned} &\pi(h(r_v) + r_v)^2 - \pi r_v^2 = \pi(h_0 + r_0)^2 - \pi r_0^2 \\ \Rightarrow\quad &h(r_v)^2 + 2r_v h(r_v) - (h_0^2 + 2r_0 h_0) = 0 \end{aligned} \tag{A24}$$

which can be solved analytically by the quadratic formula (omitting the negative root):

$$\begin{aligned} h(r_v) &= \frac{-2r_v + \sqrt{4r_v^2 + 4((r_0 + h_0)^2 - r_0^2)}}{2} \\ &= -r_v + \sqrt{r_v^2 + 2h_0 r_0 + h_0^2}. \end{aligned} \tag{A25}$$

APPENDIX B: FULL-SCALE TREE MODEL

When solving for blood flow in the full-scale vascular tree model, given a set of afferent arteriolar radii, boundary conditions must be defined to solve *Eq. 17*. The arterial pressure, P_{art}, represents the input pressure, P_{int}, to the vascular tree. The outlet boundary condition is more complex since the pressure at the end of the afferent arteriole remains unknown before the system is solved. To define the outlet boundary condition, the afferent arteriole is extended with two parts in series without branches: the efferent arteriole and the postglomerular capillaries, as shown in Fig. B1.

The pressure at the end of the postglomerular capillaries is assumed to be 0 mmHg, which defines the outlet boundary condition: $P_{PC} = 0$ mmHg, corresponding to the venous pressure. The resistances for the efferent arterioles and postglomerular capillaries (R_E and R_{PC}) are assumed to be constant. Since the three parts are connected in series, for each afferent arteriole (i,j), its equivalent total resistance is

$$R_t = R_{i,j} + R_E + R_{PC}. \tag{B1}$$

The blood flows in the afferent and efferent arterioles are not the same due to glomerular filtration. The single nephron glomerular filtration rate, $Q^T_{i,j}(0)$, is one of the results of the nephron model. To solve the model, the three parts are first regarded as one vessel with equivalent flow Q_t and resistance R_t as shown in Fig. B1. The real flow in the afferent arteriole ($Q_{i,j}$) is then recovered by the properties $Q_E = Q_{i,j} - Q^T_{i,j}(0)$ and $Q_{PC} = Q_{i,j}$. In the first round, $Q^T_{i,j}(0)$ is assumed to be 1/3 of the plasma flow, where the plasma flow is assumed to be half of the blood flow from the afferent arteriole. Therefore, we assume $Q^T_{i,j}(0) = \frac{Q_{i,j}}{6}$ in the first round. From the second round, $Q^T_{i,j}(0)$ will be calculated from the nephron model by *Eq. 1*.

Given the above boundary conditions, the system of linear equations is given as:

$$\begin{cases} P_n = P_{art}, & n = n_{in} \\ P_n = 0, & n \in N_{terminal} \\ \left(\frac{1}{R_{p,n}} + \sum_{i \in B_v/p} \frac{1}{R_{n,i}}\right) P_n - \left(\frac{P_p}{R_{p,n}} + \sum_{i \in B_v/p} \frac{P_i}{R_{n,i}}\right) = 0, & \text{otherwise}, \end{cases} \tag{B2}$$

where n_{in} denotes the single root node and $N_{terminal}$ denotes the set of terminal nodes of the vascular tree. The resistance term in *Eq. B2* is defined as

$$R_{i,j} = \begin{cases} \frac{8\mu l_{i,j}}{\pi r_{i,j}^4}, & j \notin N_{terminal} \\ \frac{8\mu l_{i,j}}{\pi r_{i,j}^4} + R_E + R_{PC}, & \text{otherwise}, \end{cases} \tag{B3}$$

where R_E and R_{PC} are constant.

The system of linear equations defined in *Eq. B2* can be expressed in matrix form

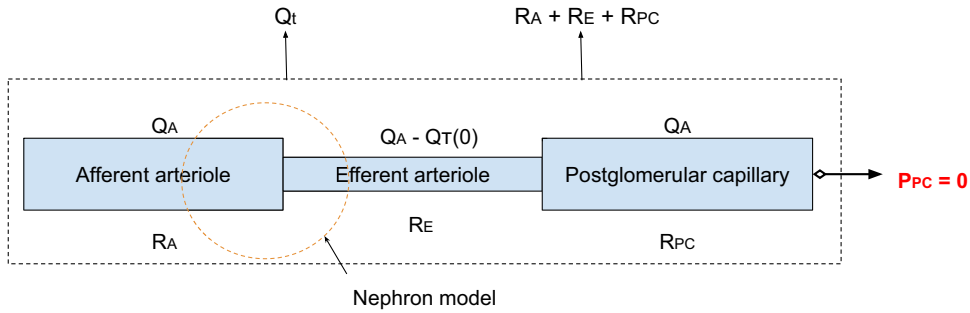


Figure B1. Extended vascular model showing the afferent and efferent arterioles, the post glomerular capillary bed, and their associated blood flows and resistances, together with the nephron model.

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$$AV = b, \tag{B4}$$

where V is the column of pressure at every node. The values in the sparse matrix A and the vector b are determined by *Eq. B2*. V can then be solved using standard methods, such as Gaussian elimination.

The previous equation solves for the pressure P_n at every node n, allowing the calculation of flow Q_e in every vessel segment $e = (i,j)$ by Poiseuille's law

$$Q_e = \frac{P_i - P_j}{R_e}. \tag{B5}$$

For $j \in N_{terminal}$, the flow Q_t computed by Kirchhoff's law represents the equivalent flow of the two parts. The real flow $Q_{i,j}$ associated with the afferent arteriole (i,j) can be recovered by the equal pressure drop property

$$\begin{aligned} Q_t(R_{i,j} + R_E + R_{PC}) &= Q_{i,j}R_{i,j} + (Q_{i,j} - Q^T_{i,j}(0))R_E + Q_{i,j}R_{PC} \\ \Rightarrow Q_{i,j} &= Q_t + Q^T_{i,j}(0)\frac{R_E}{R_{i,j} + R_E + R_{PC}}. \end{aligned} \tag{B6}$$

APPENDIX C: SOFTWARE IMPLEMENTATION

The model is implemented in Python version 3.10 using the packages NumPy and SciPy. The model of the full-scale vascular tree is implemented as a directed graph using NetworkX (74). All ordinary differential equations are solved using the *solve_ivp* method in SciPy by the implicit Runge–Kutta method of the Radau IIA family of order 5 (75). For the algebraic equations, root finding is performed numerically using the *fsolve* routine in SciPy that implements MINPACK's hybrid algorithm, which combines the Newton–Raphson method and quasi-Newton methods (76).

The software was run in Ubuntu 22.04 with an Intel Core i7-8700 processor at 3.20 GHz and 24 GB RAM. The Intel Core i7-8700 processor has 6 cores and supports up to 12 simultaneous threads. Since in the current model, each nephron acts as an independent unit, the models for the individual nephrons were solved in parallel using the built-in concurrent module in Python. When solving the model at a given renal arterial pressure, it takes approximately half an hour for one iteration of the entire model and ~20 iterations are necessary for the renal blood flow to reach convergence.

DATA AVAILABILITY

The full-scale renal arterial tree model (in .vtk format) with the attached nephron model is given in the GitHub repository https://github.com/Paddy-Xu/RenalFullSimu. Detailed simulation results can be made available upon request to the corresponding author. The source code for the model is available at GitHub repository https://github.com/Paddy-Xu/RenalFullSimu under Apache License 2.0 open-source license.

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DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the authors.

AUTHOR CONTRIBUTIONS

P.X., S.D., O.S., and N.-H.H.-R. conceived and designed research; P.X. performed experiments; P.X., O.S., and N.-H.H.-R. analyzed data; P.X., O.S., and N.-H.H.-R. interpreted results of experiments; P.X., O.S., and N.-H.H.-R. prepared figures; P.X., O.S., and N.-H.H.-R. drafted manuscript; P.X., S.D., O.S., and N.-H.H.-R. edited and revised manuscript; P.X., S.D., O.S., and N.-H.H.-R. approved final version of manuscript.

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